Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-12 20:35:45

Data & configuration

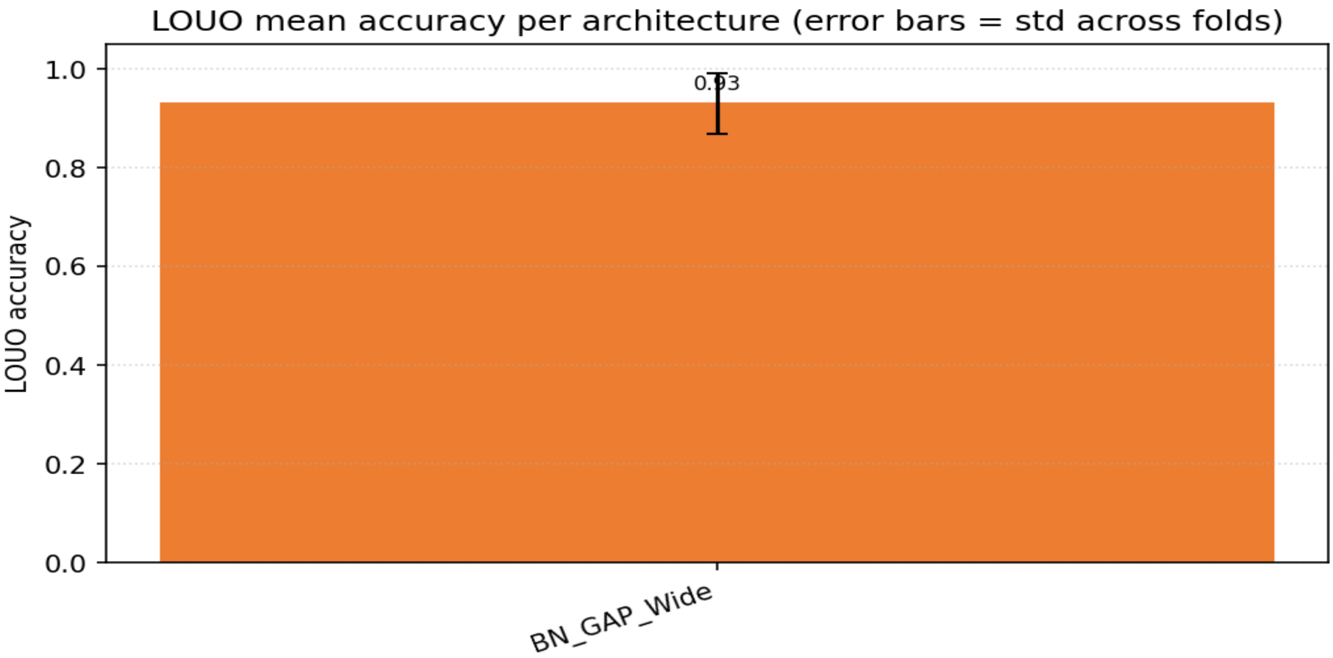
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	— (none)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	off
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	BN_GAP_Wide
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

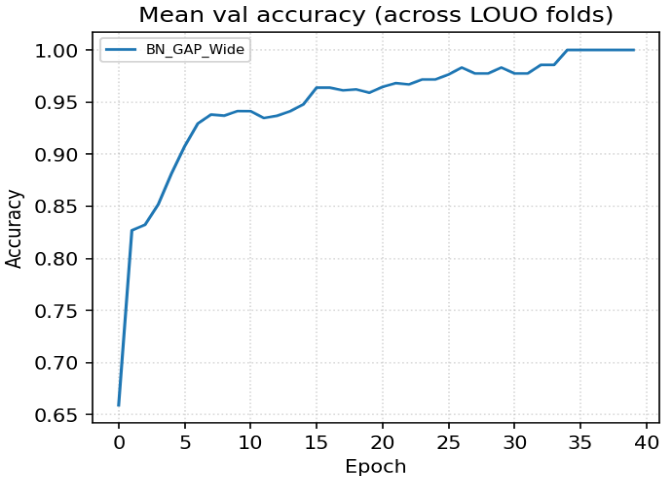
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP_Wide	104,423	0.9292	0.0614	13	—

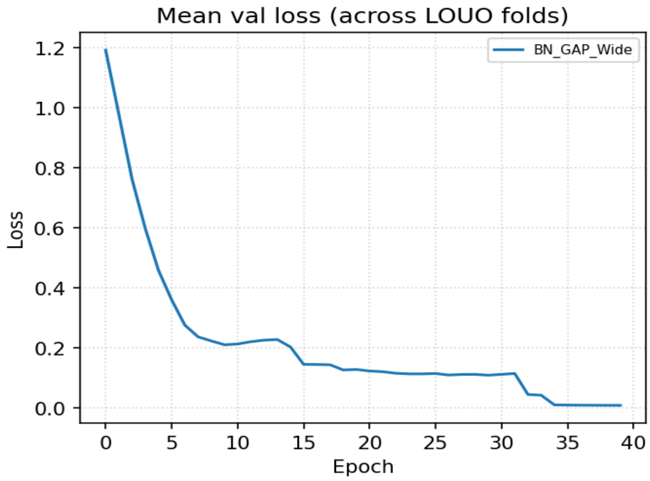
Best LOUO mean: BN_GAP_Wide

LOUO accuracy per variant



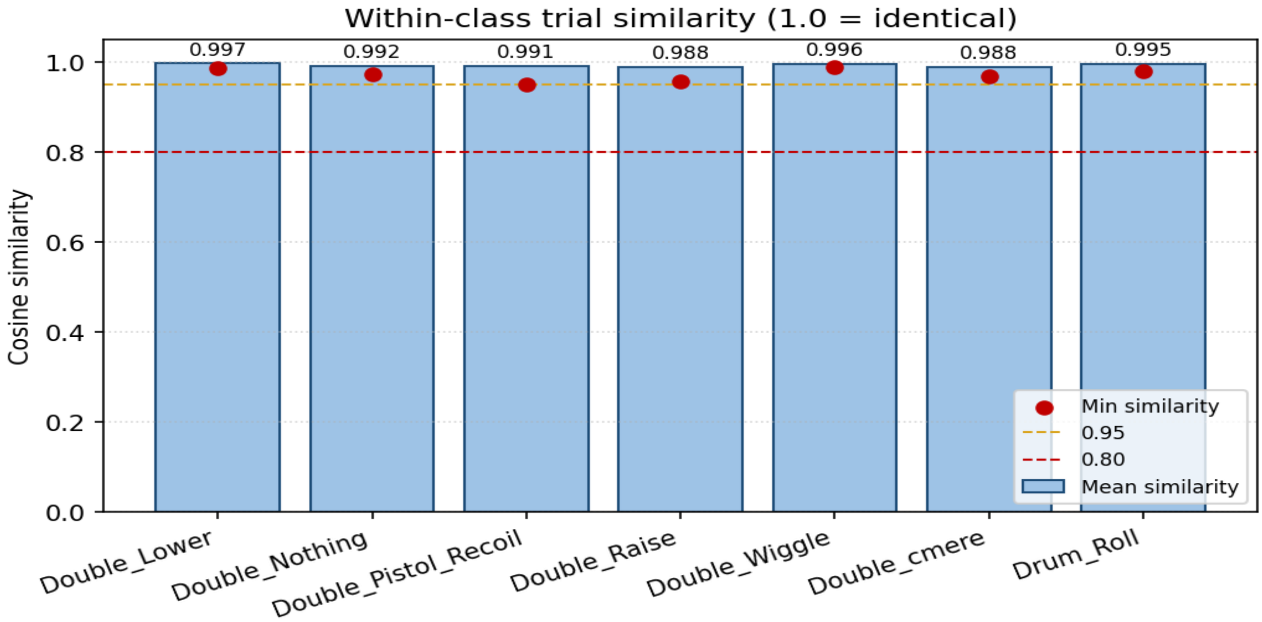
Mean training curves across LOUO folds

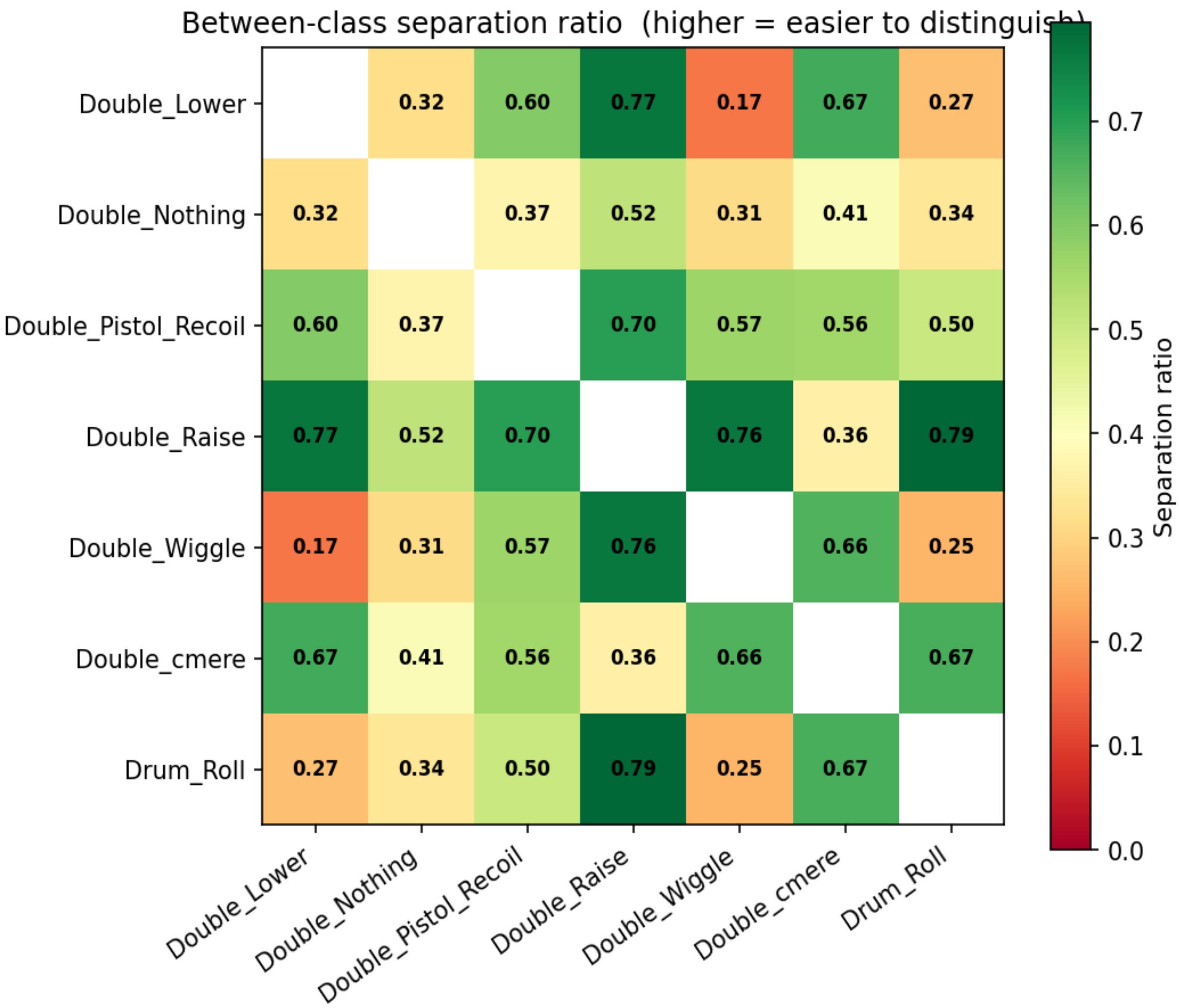


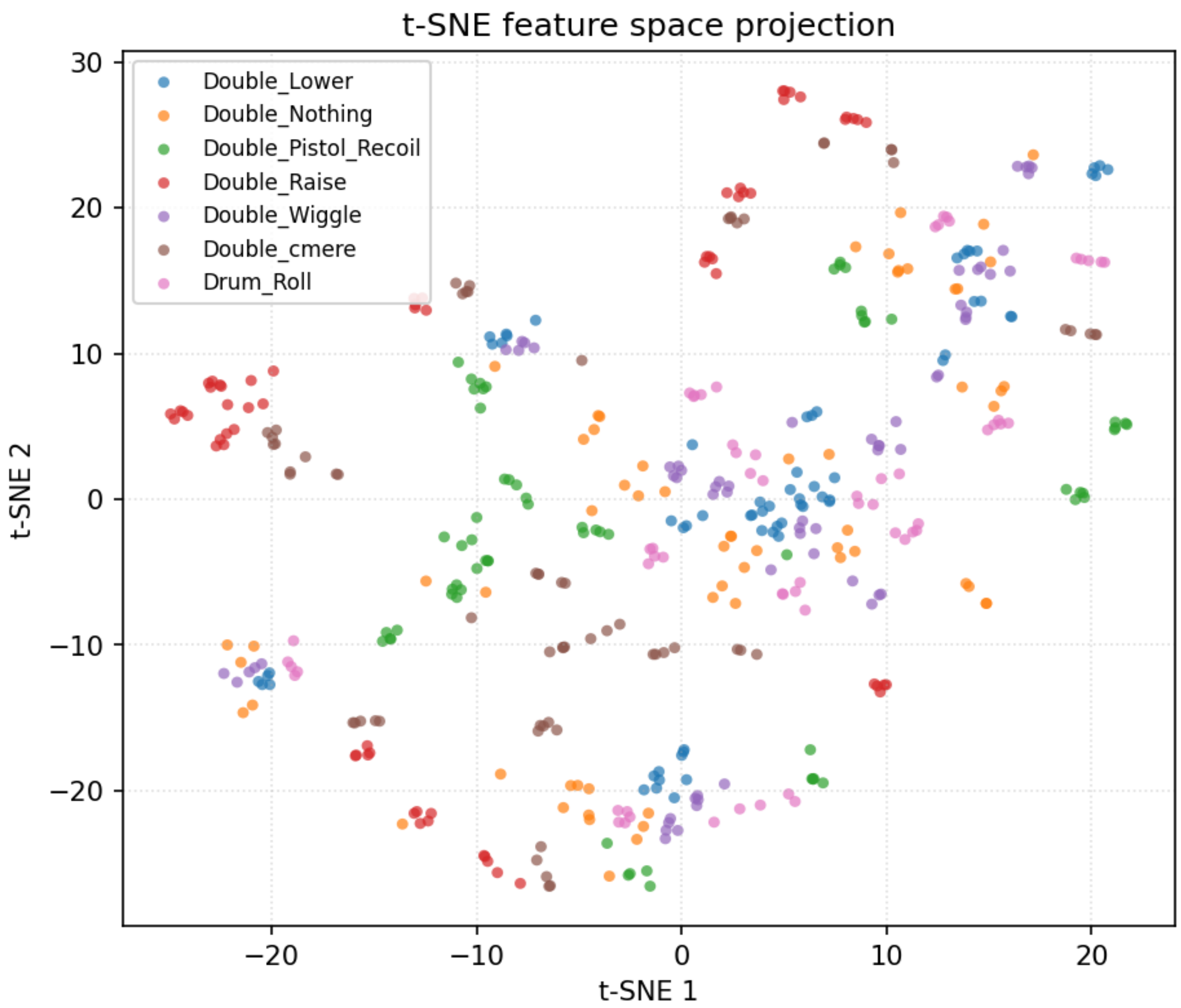


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



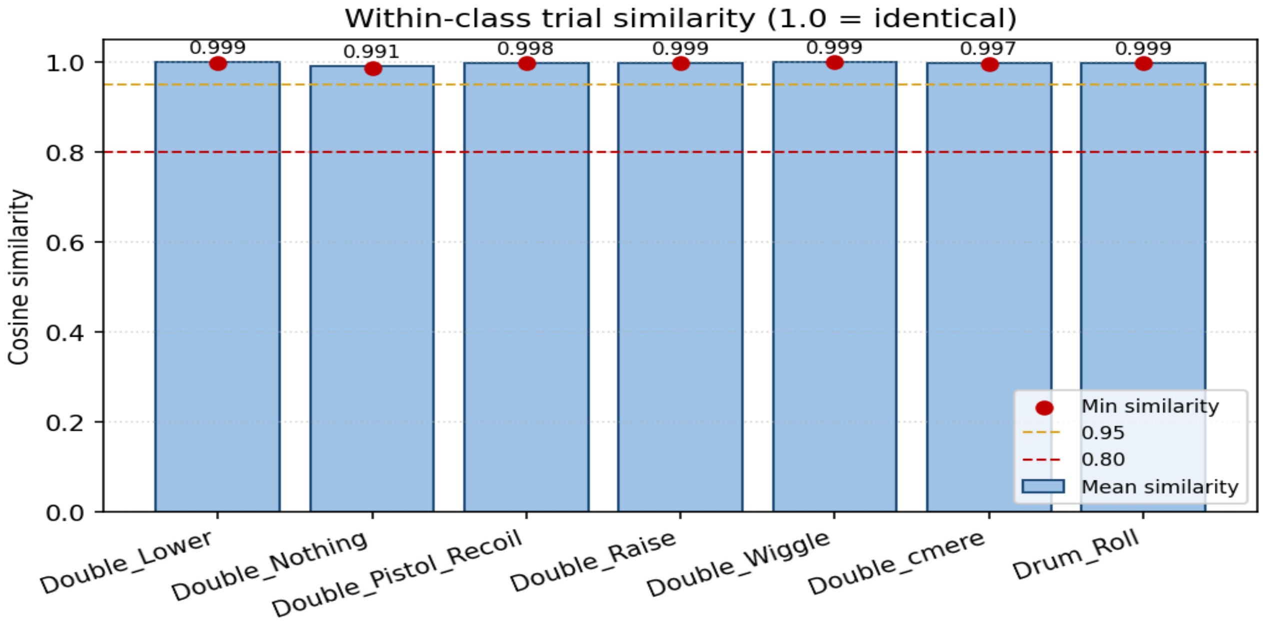


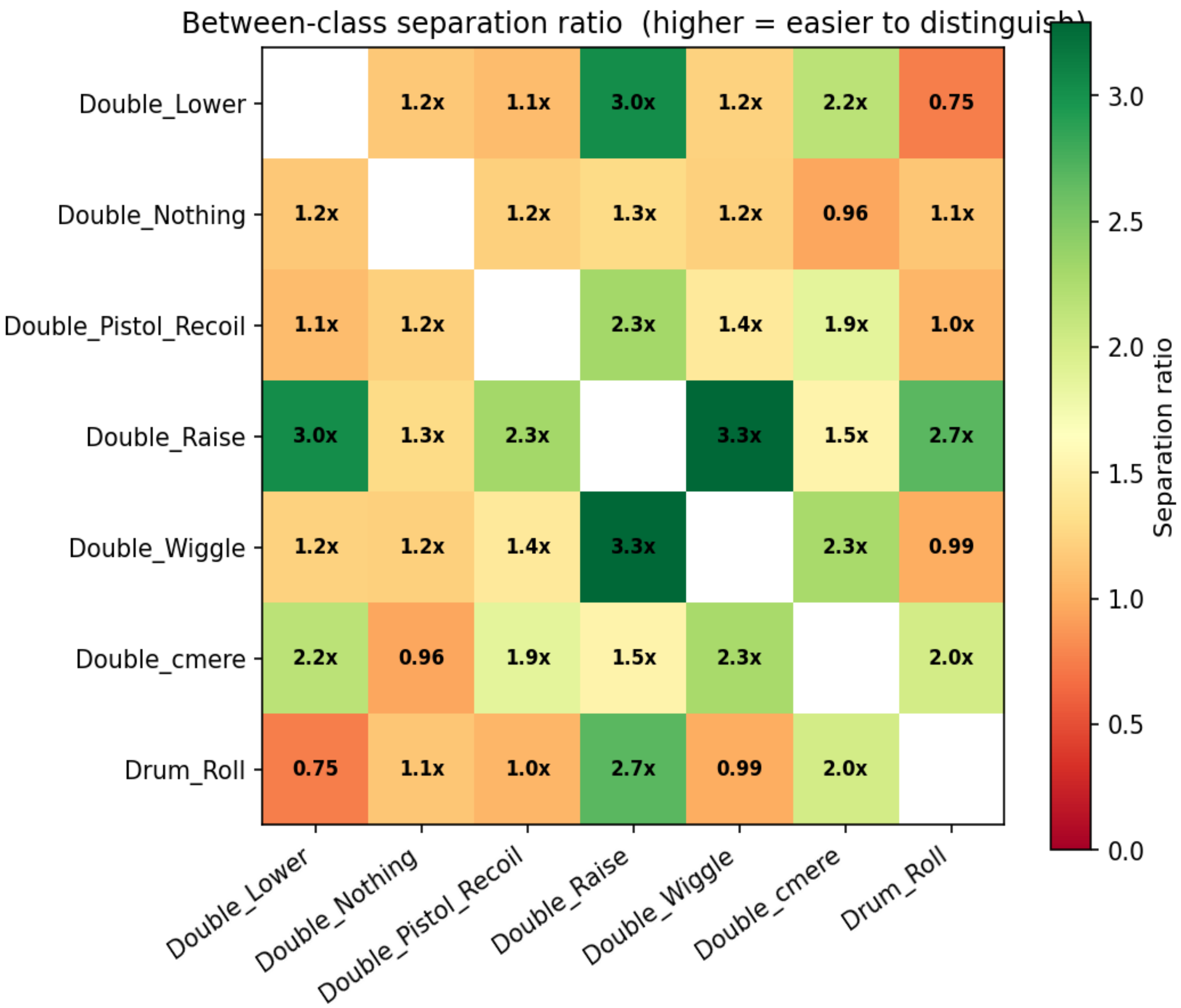


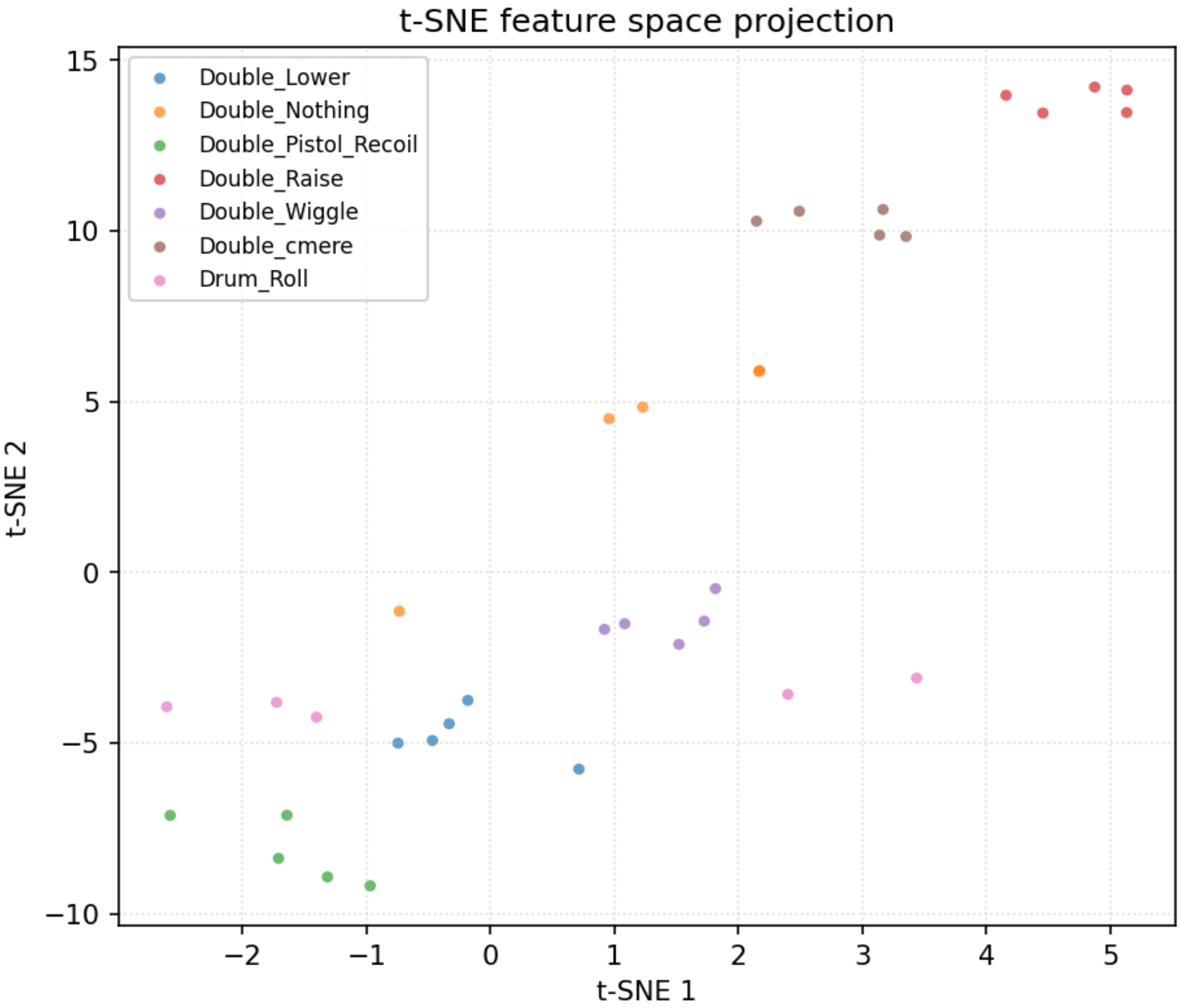
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

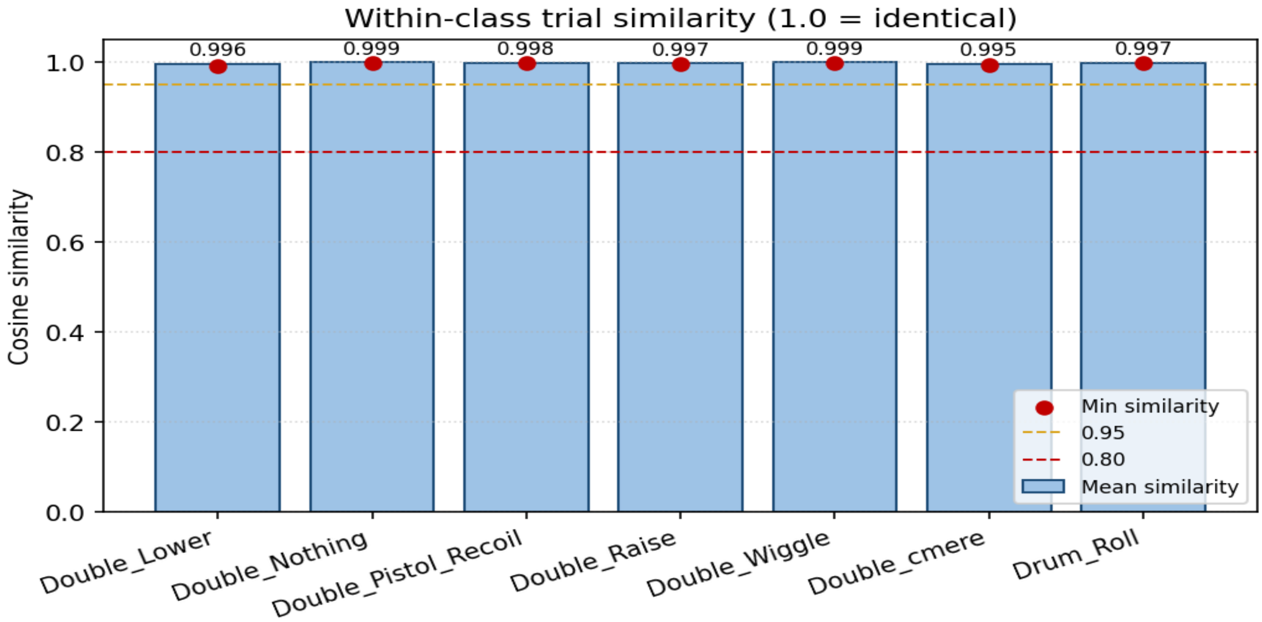


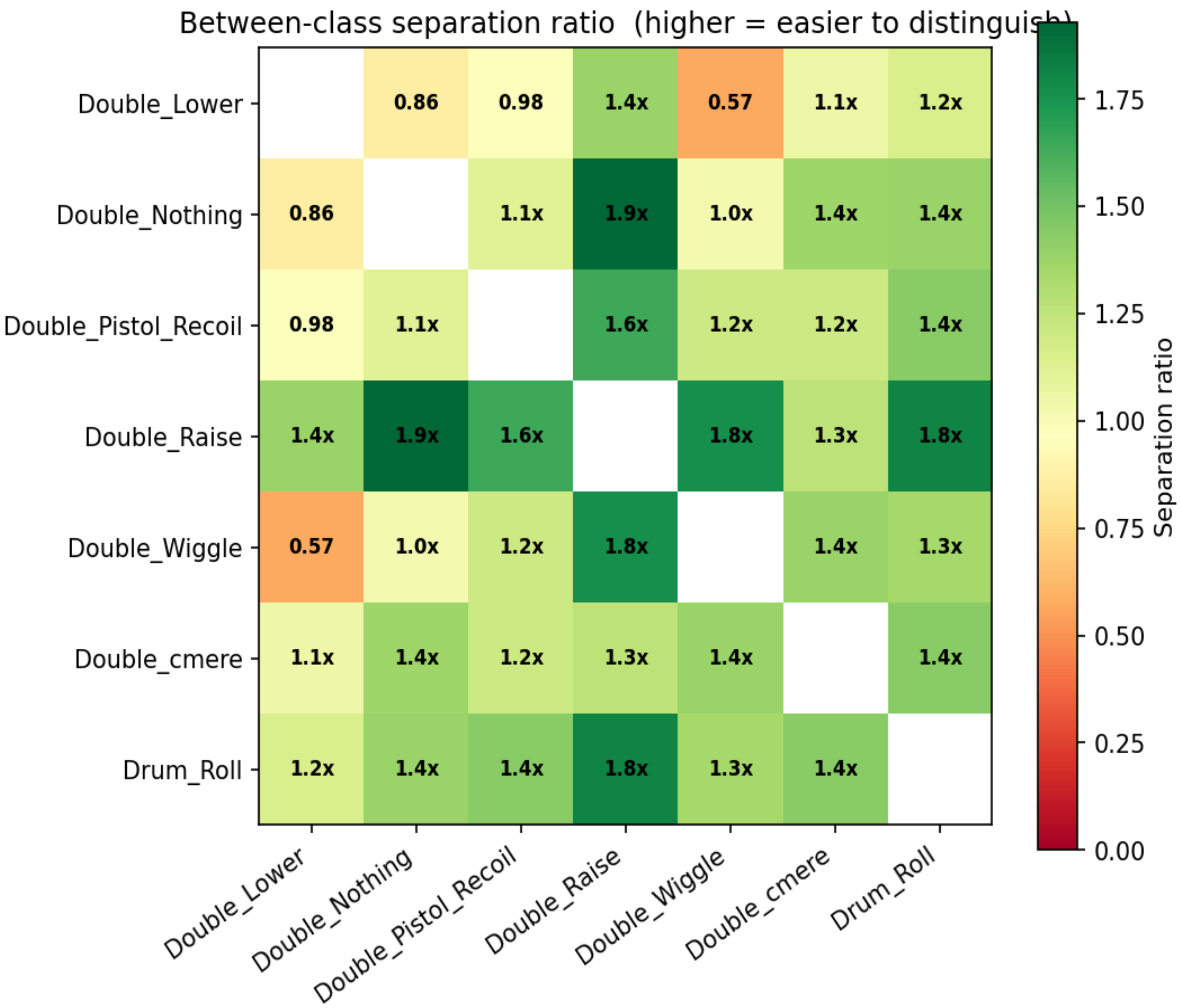



t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 1
t-SNE 2
15
10
5
0
−5
−10
−2
−1
0
1
2
3
4
5

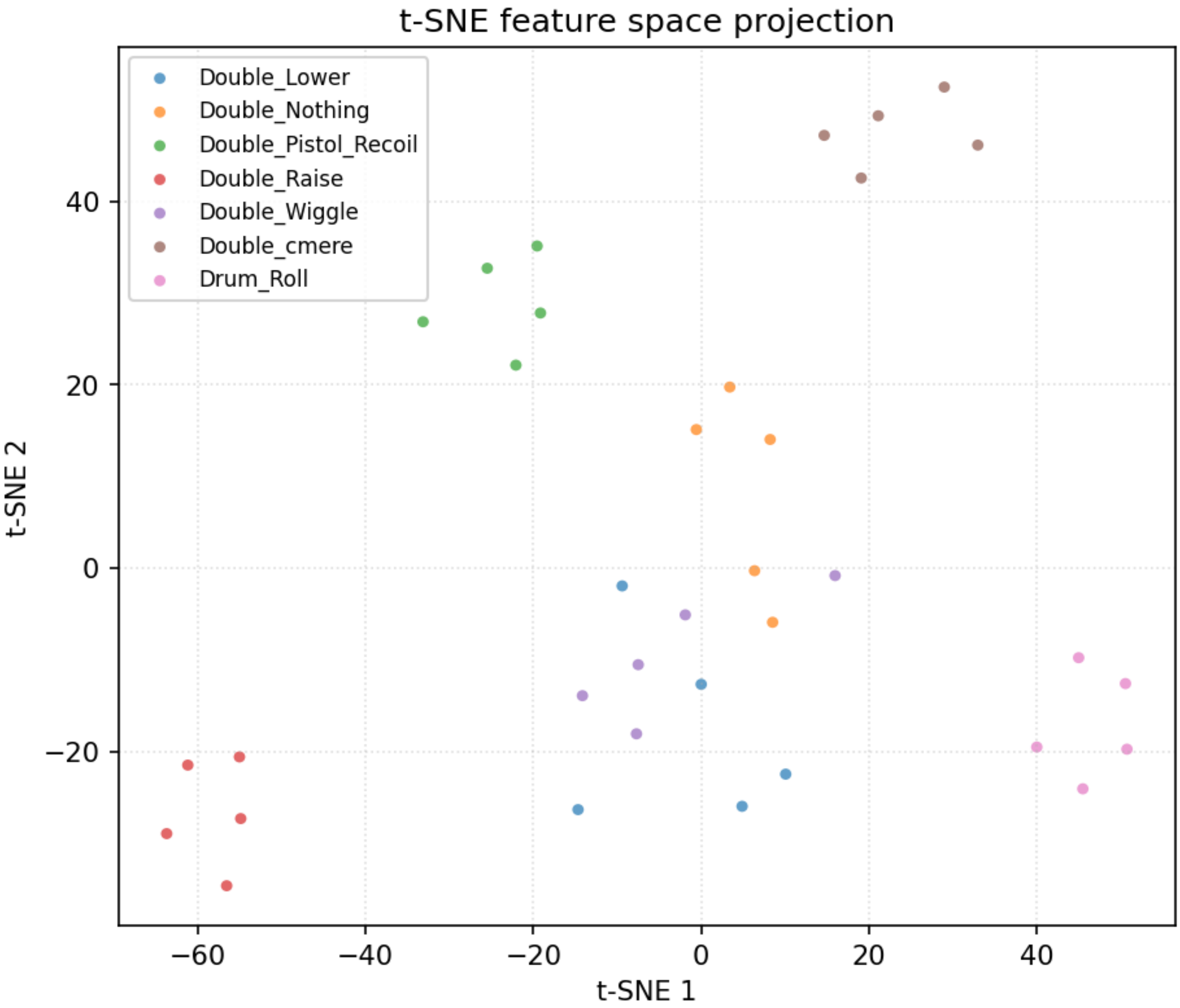
Data diagnostics — user 12_Marcus

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).



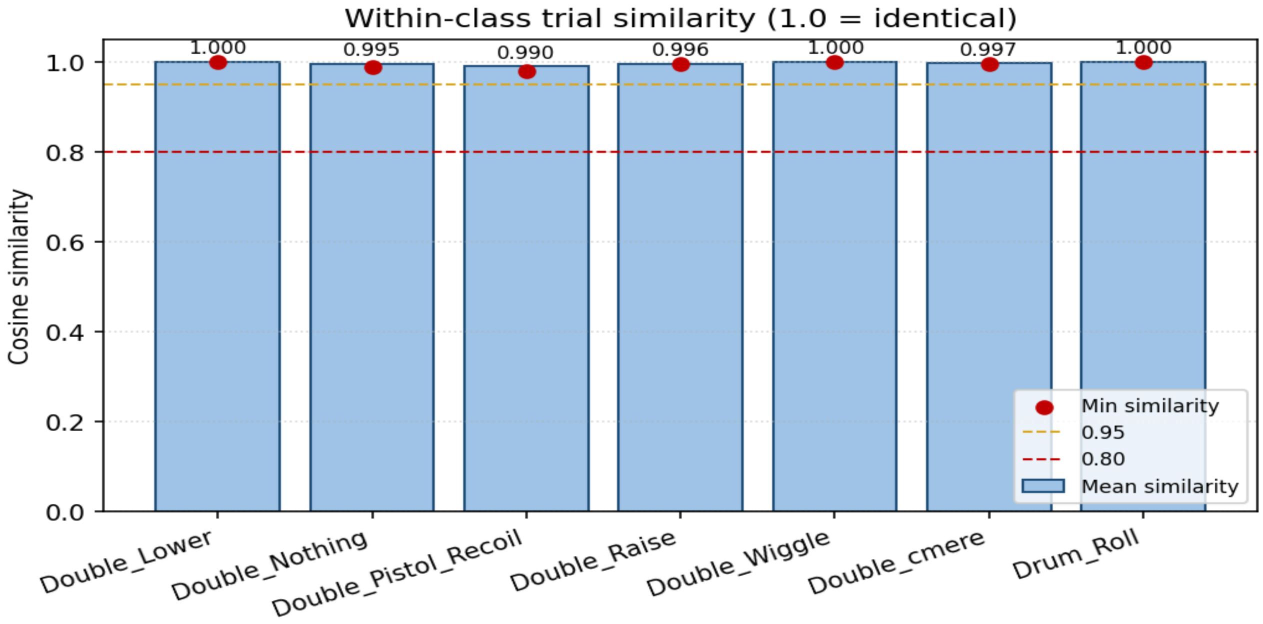


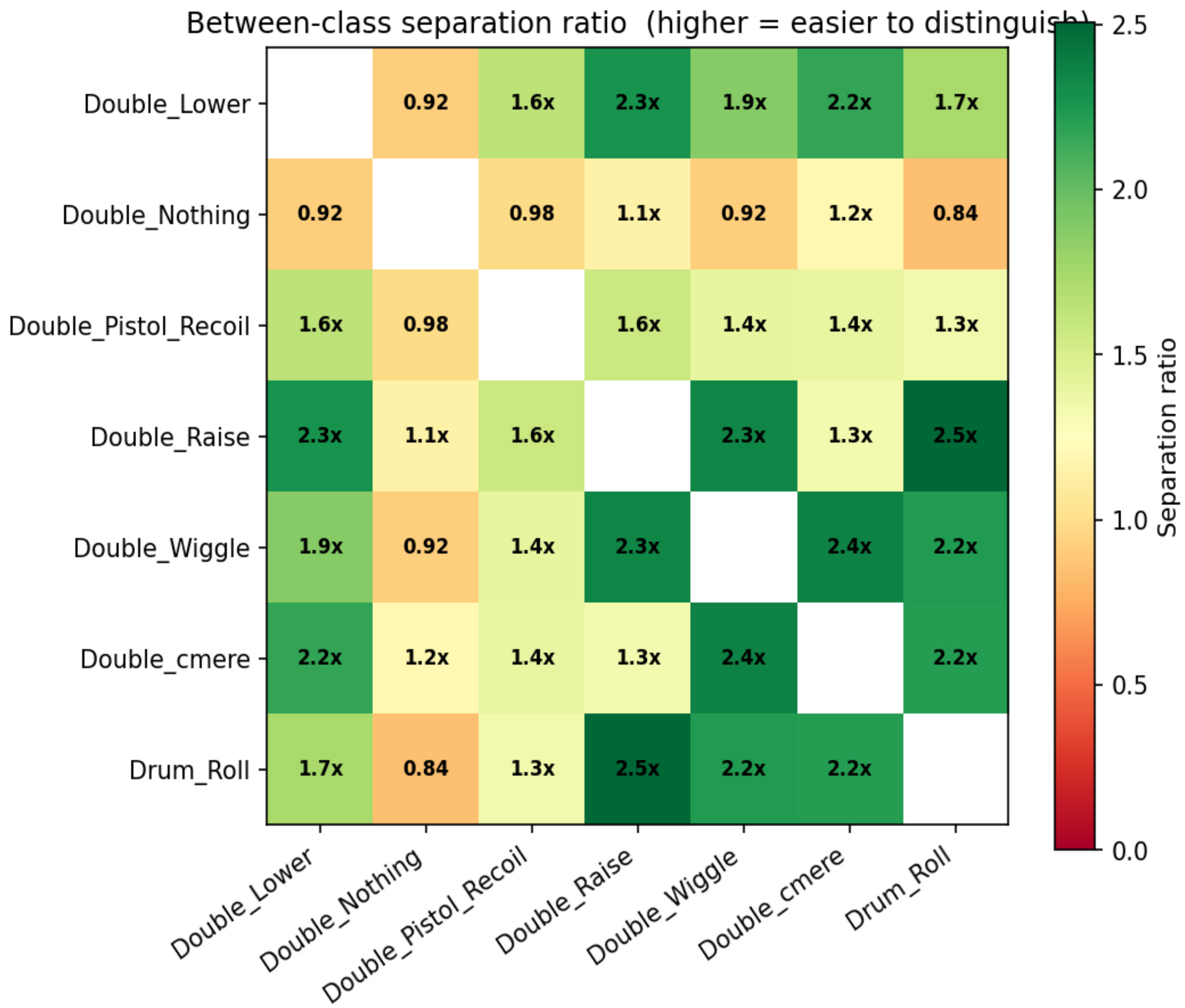
	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		0.86	0.98	1.4x	0.57	1.1x	1.2x
Double_Nothing	0.86		1.1x	1.9x	1.0x	1.4x	1.4x
Double_Pistol_Recoil	0.98	1.1x		1.6x	1.2x	1.2x	1.4x
Double_Raise	1.4x	1.9x	1.6x		1.8x	1.3x	1.8x
Double_Wiggle	0.57	1.0x	1.2x	1.8x		1.4x	1.3x
Double_cmere	1.1x	1.4x	1.2x	1.3x	1.4x		1.4x
Drum_Roll	1.2x	1.4x	1.4x	1.8x	1.3x	1.4x	

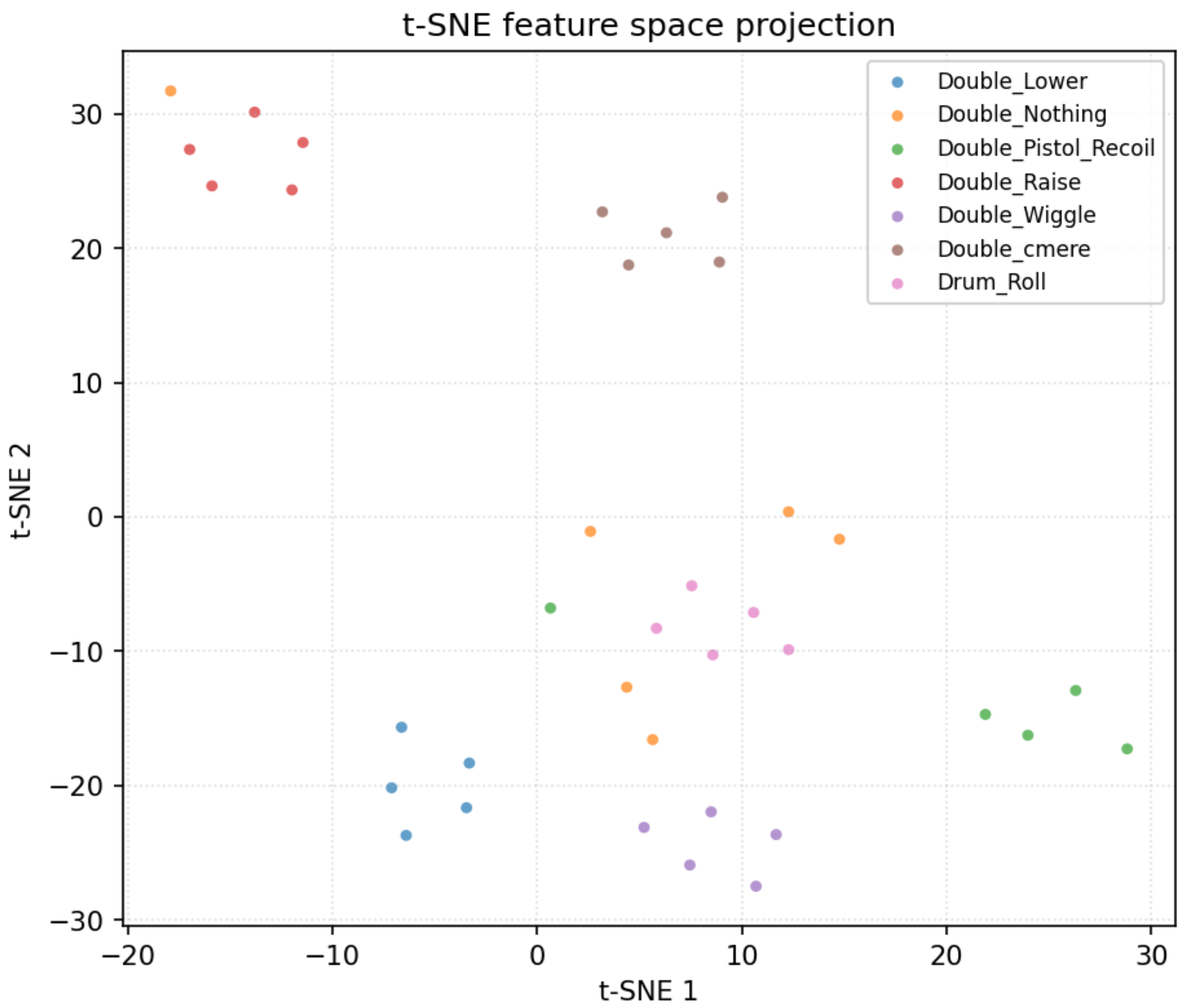

t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
40
20
0
−20
−60
−40
−20
0
20
40
t-SNE 1
t-SNE 2

Data diagnostics — user 14_StephenV2

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).



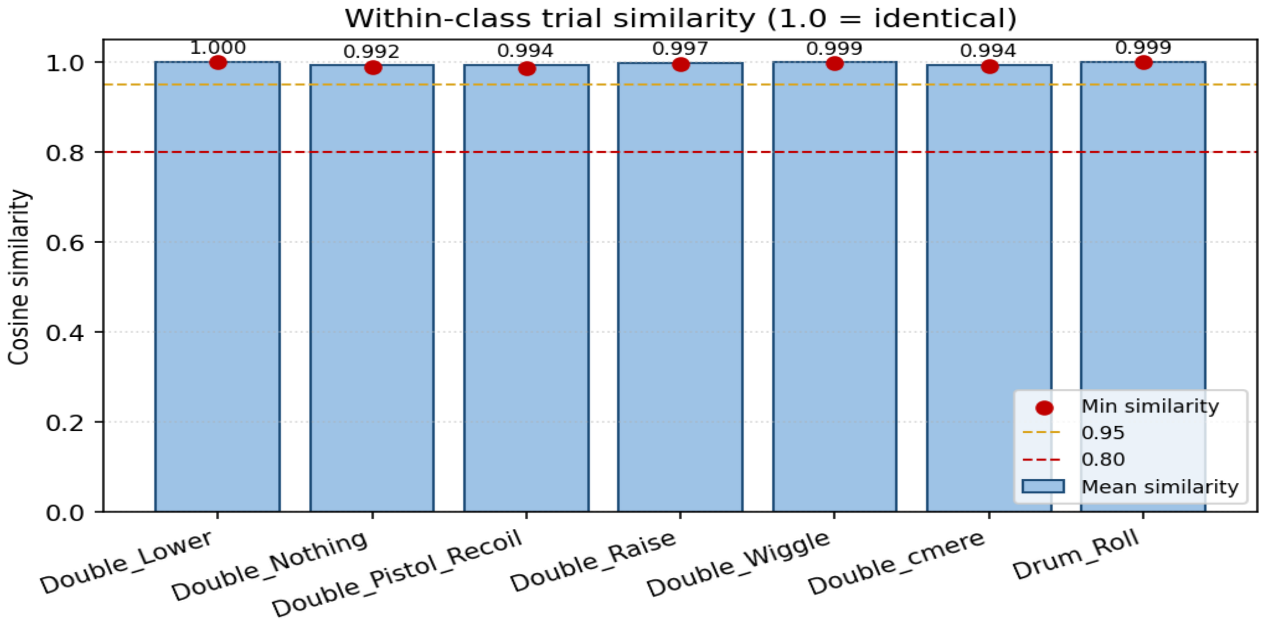


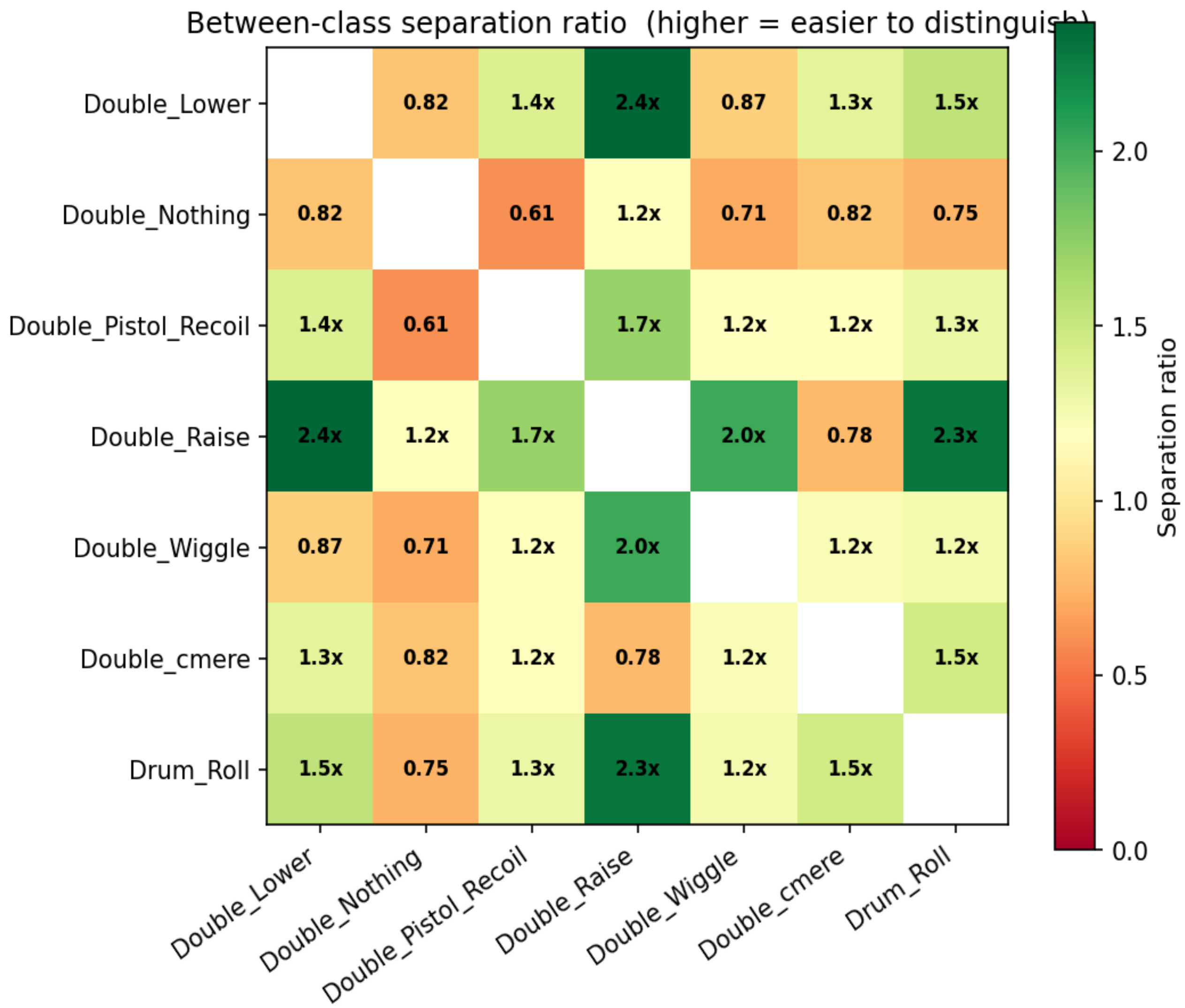


t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
30
20
10
0
−10
−20
−30
−20
−10
0
10
20
30

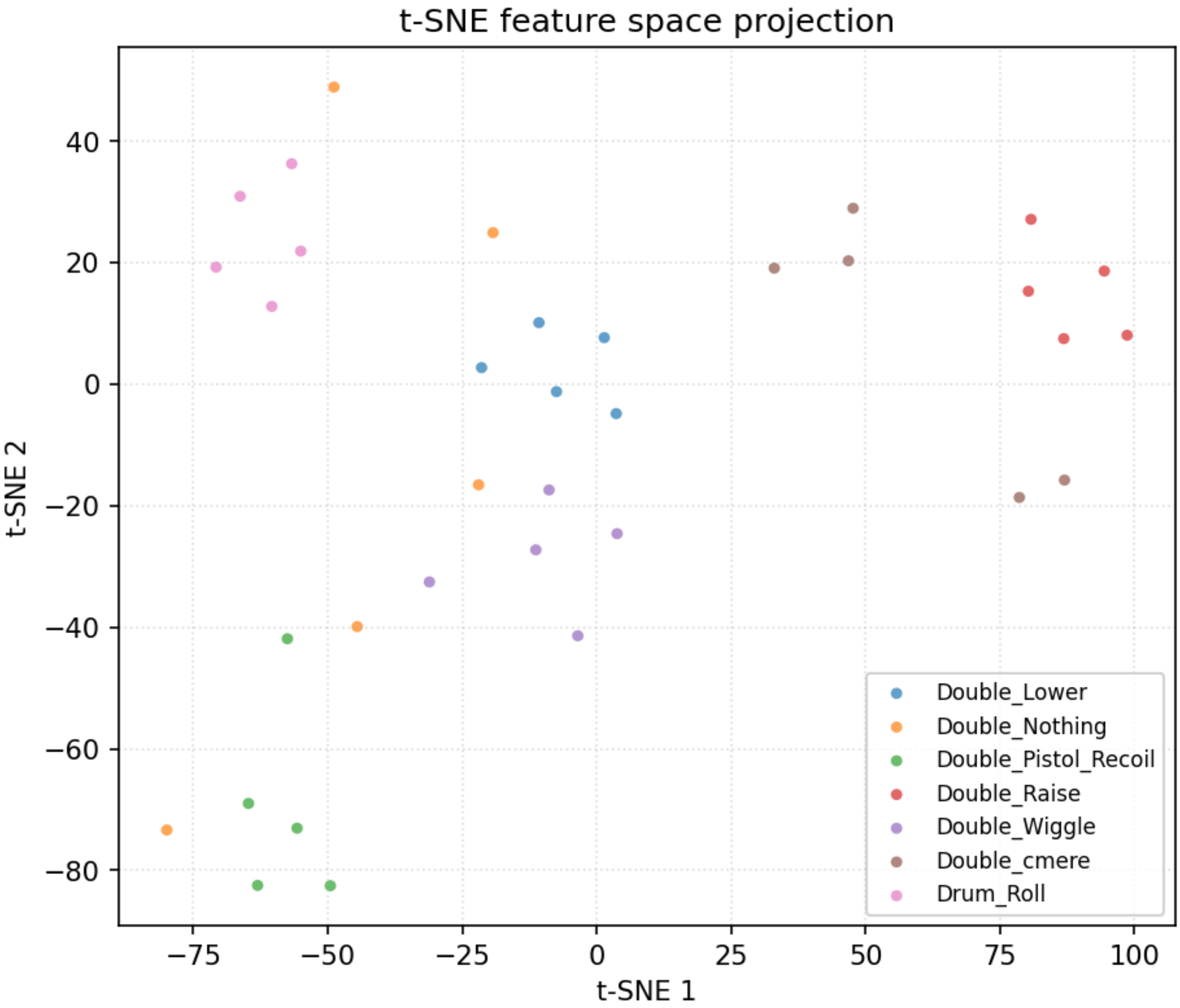
Data diagnostics — user 15_Tash

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).





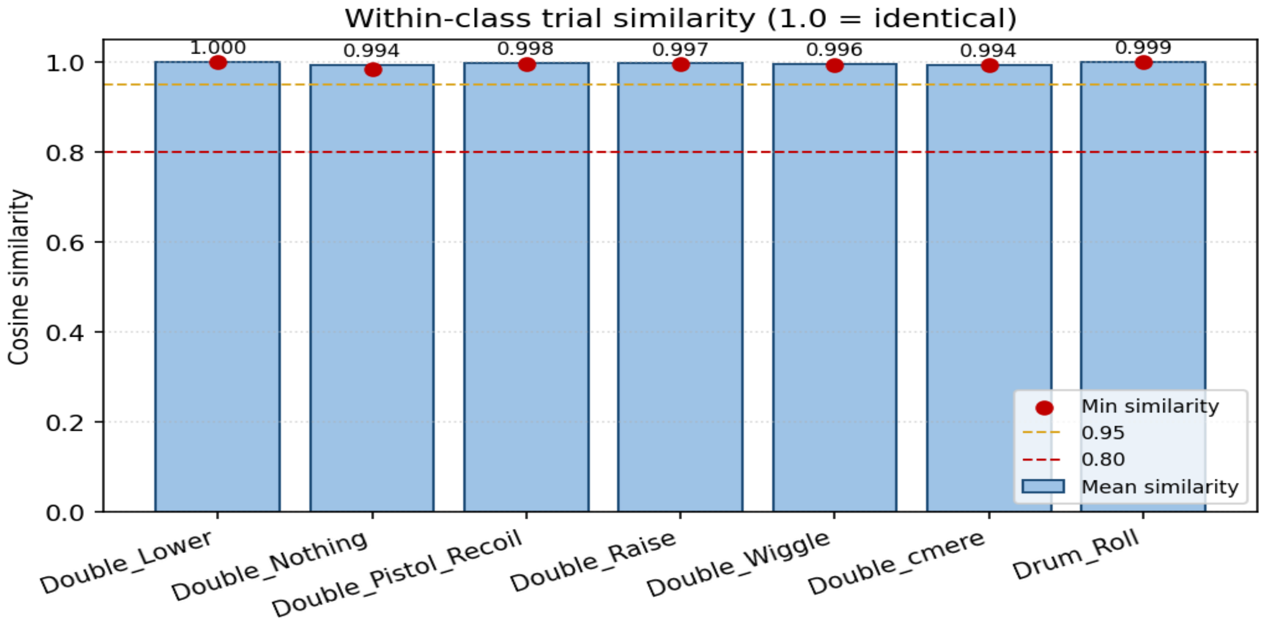
	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		0.82	1.4x	2.4x	0.87	1.3x	1.5x
Double_Nothing	0.82		0.61	1.2x	0.71	0.82	0.75
Double_Pistol_Recoil	1.4x	0.61		1.7x	1.2x	1.2x	1.3x
Double_Raise	2.4x	1.2x	1.7x		2.0x	0.78	2.3x
Double_Wiggle	0.87	0.71	1.2x	2.0x		1.2x	1.2x
Double_cmere	1.3x	0.82	1.2x	0.78	1.2x		1.5x
Drum_Roll	1.5x	0.75	1.3x	2.3x	1.2x	1.5x	

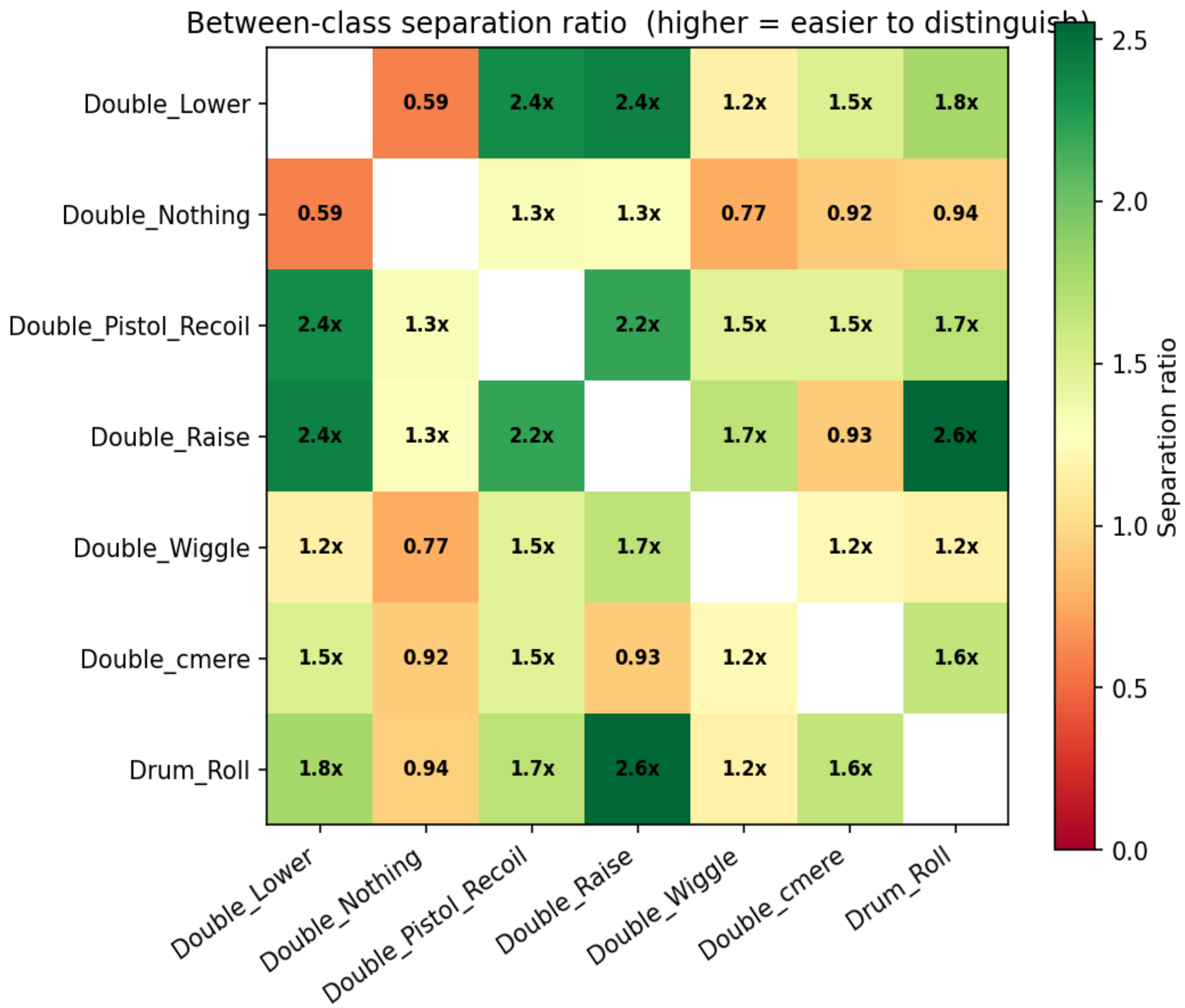


t-SNE feature space projection
t-SNE 2
t-SNE 1
40
20
0
−20
−40
−60
−80
−75
−50
−25
0
25
50
75
100
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

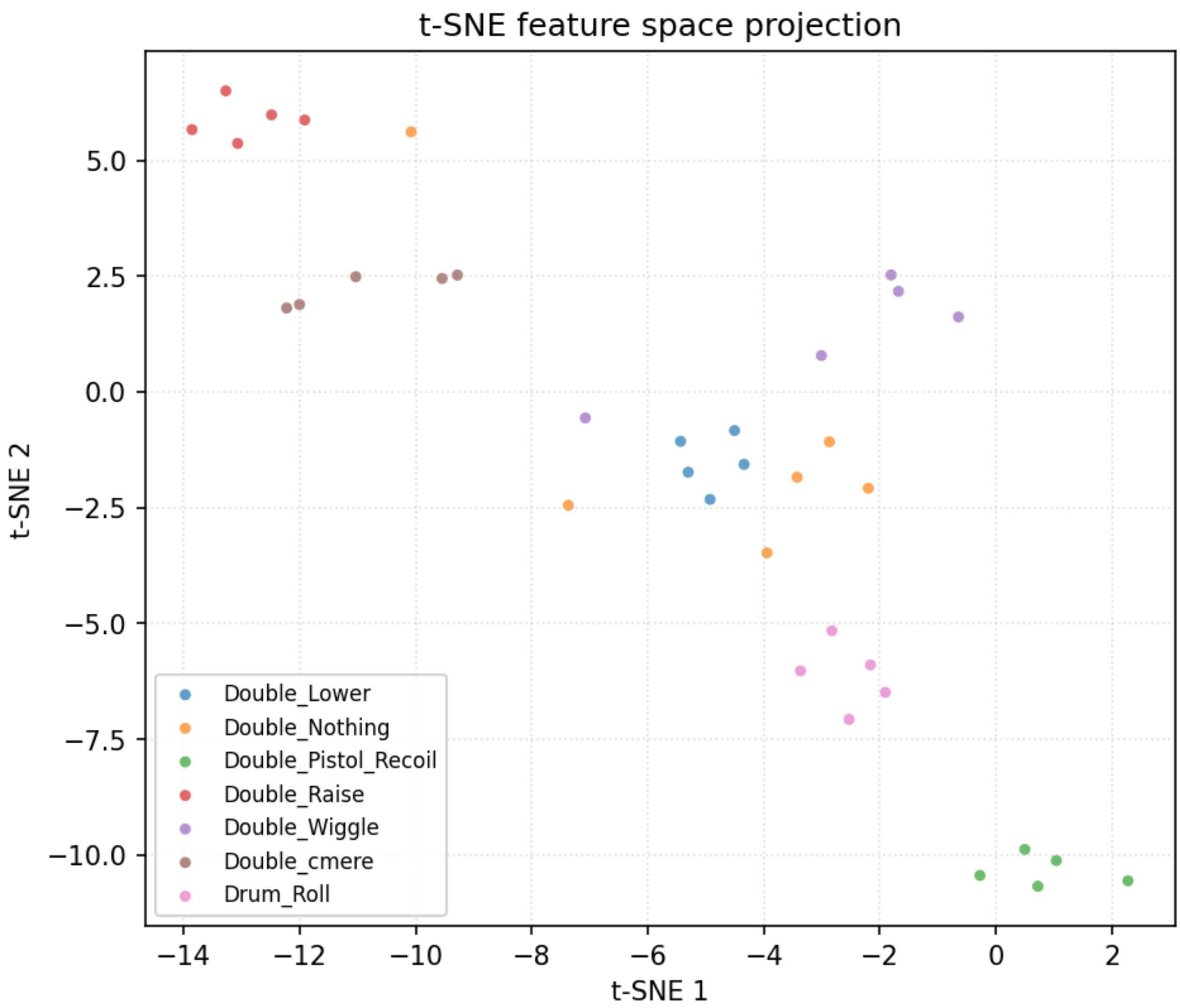
Data diagnostics — user 16_Bella

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).



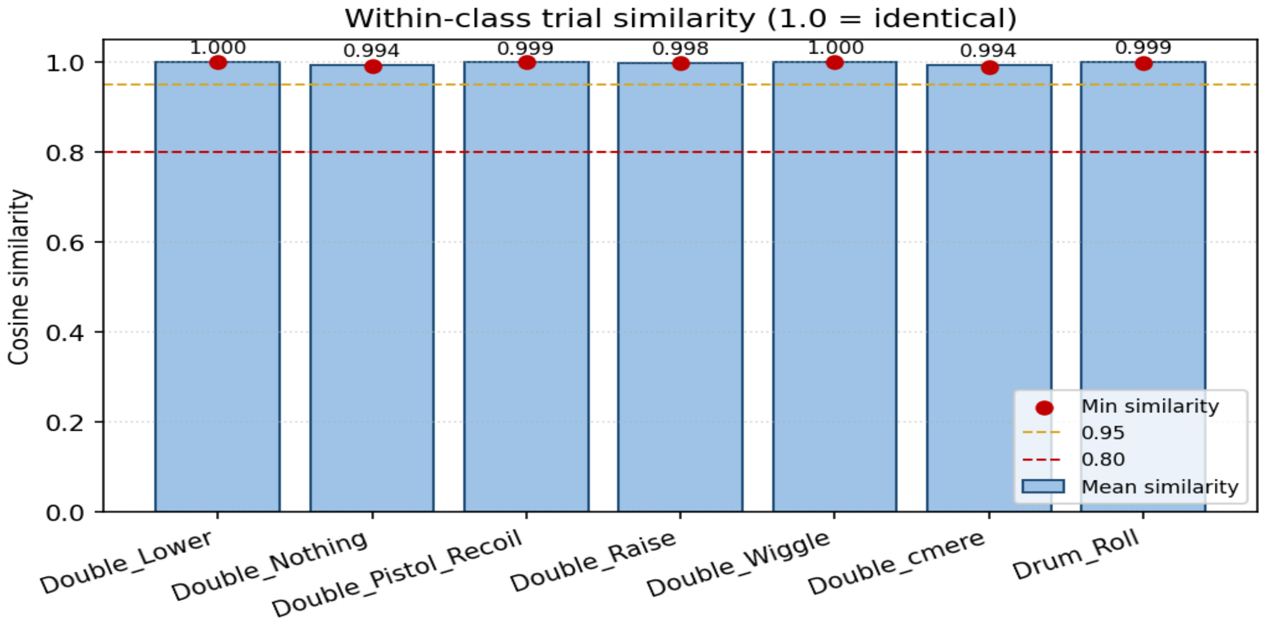


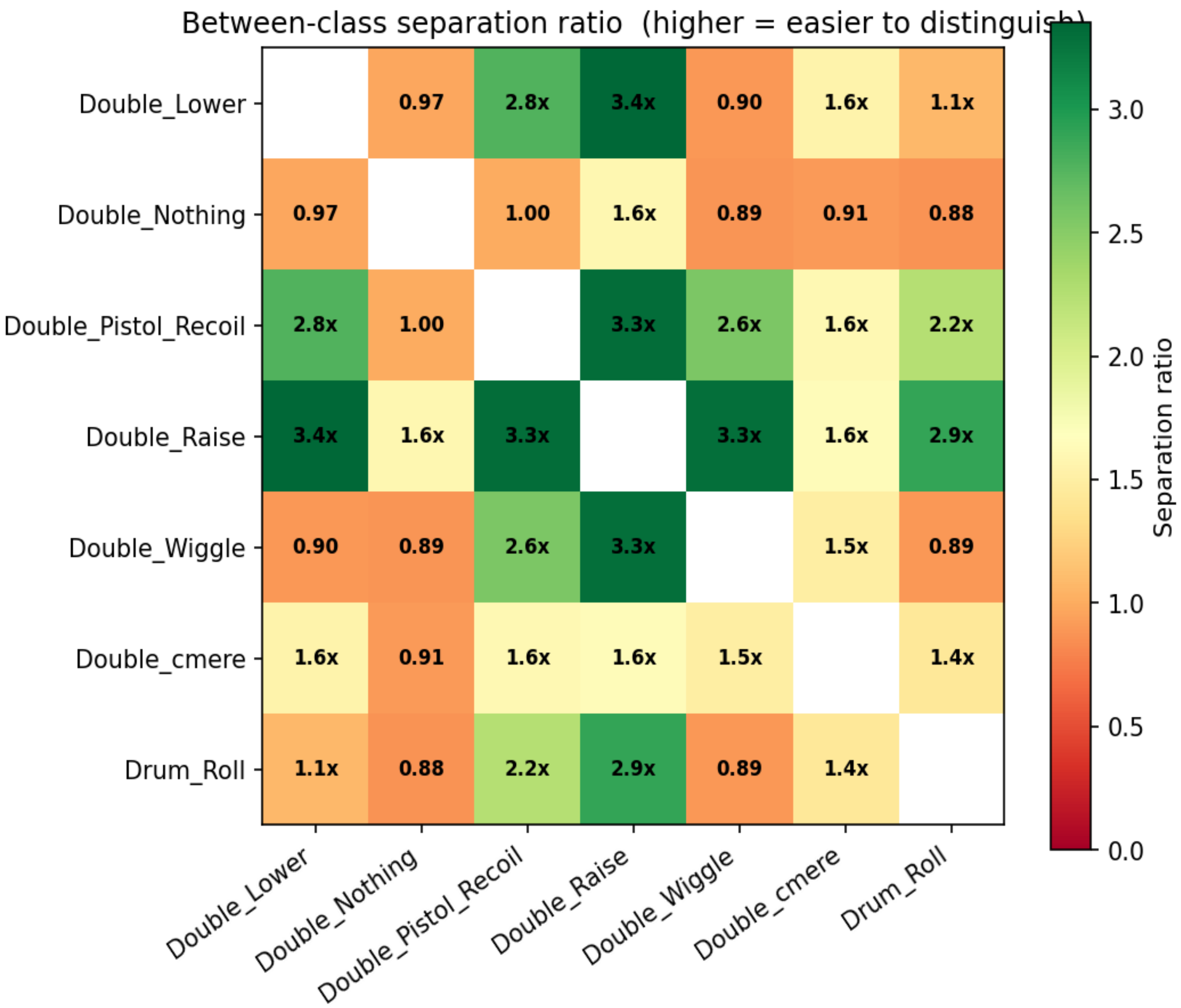
	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		0.59	2.4x	2.4x	1.2x	1.5x	1.8x
Double_Nothing	0.59		1.3x	1.3x	0.77	0.92	0.94
Double_Pistol_Recoil	2.4x	1.3x		2.2x	1.5x	1.5x	1.7x
Double_Raise	2.4x	1.3x	2.2x		1.7x	0.93	2.6x
Double_Wiggle	1.2x	0.77	1.5x	1.7x		1.2x	1.2x
Double_cmere	1.5x	0.92	1.5x	0.93	1.2x		1.6x
Drum_Roll	1.8x	0.94	1.7x	2.6x	1.2x	1.6x	

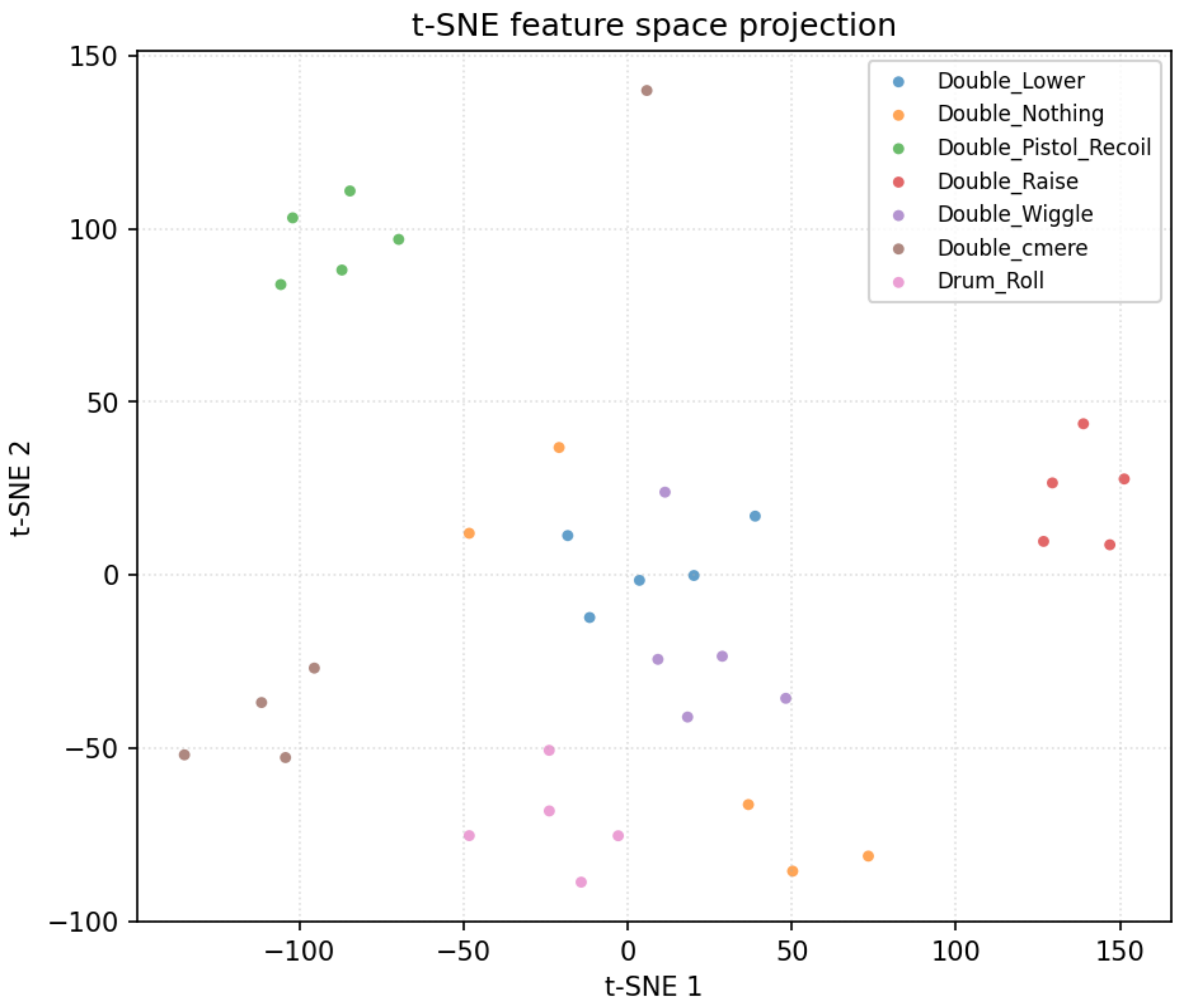

t-SNE feature space projection
t-SNE 2
t-SNE 1
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−14
−12
−10
−8
−6
−4
−2
0
2
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

Data diagnostics — user 17_Raquel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).



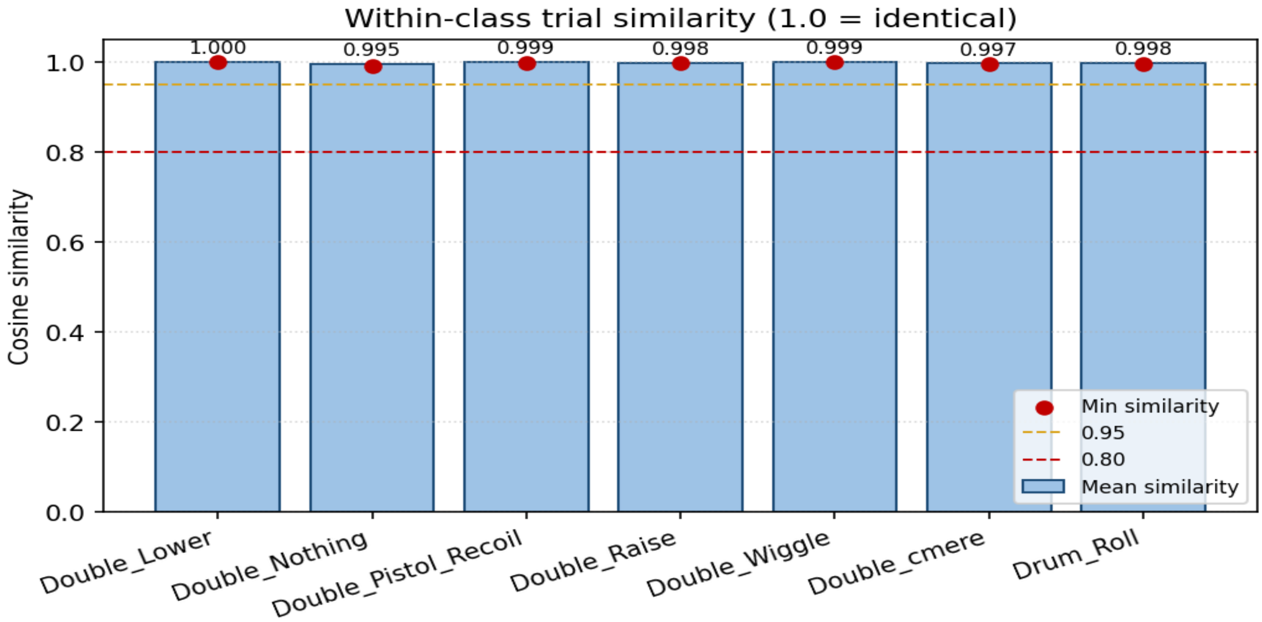


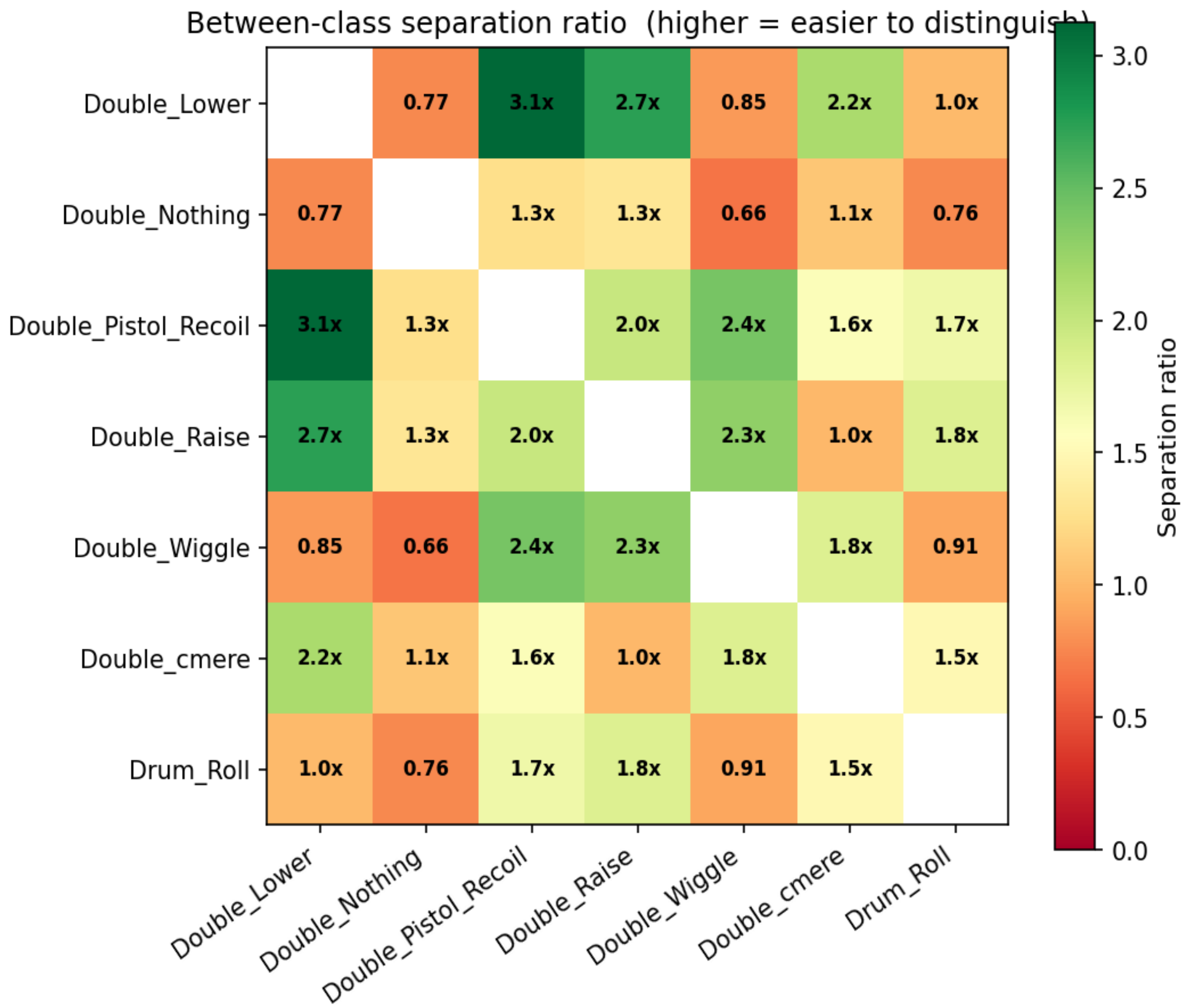


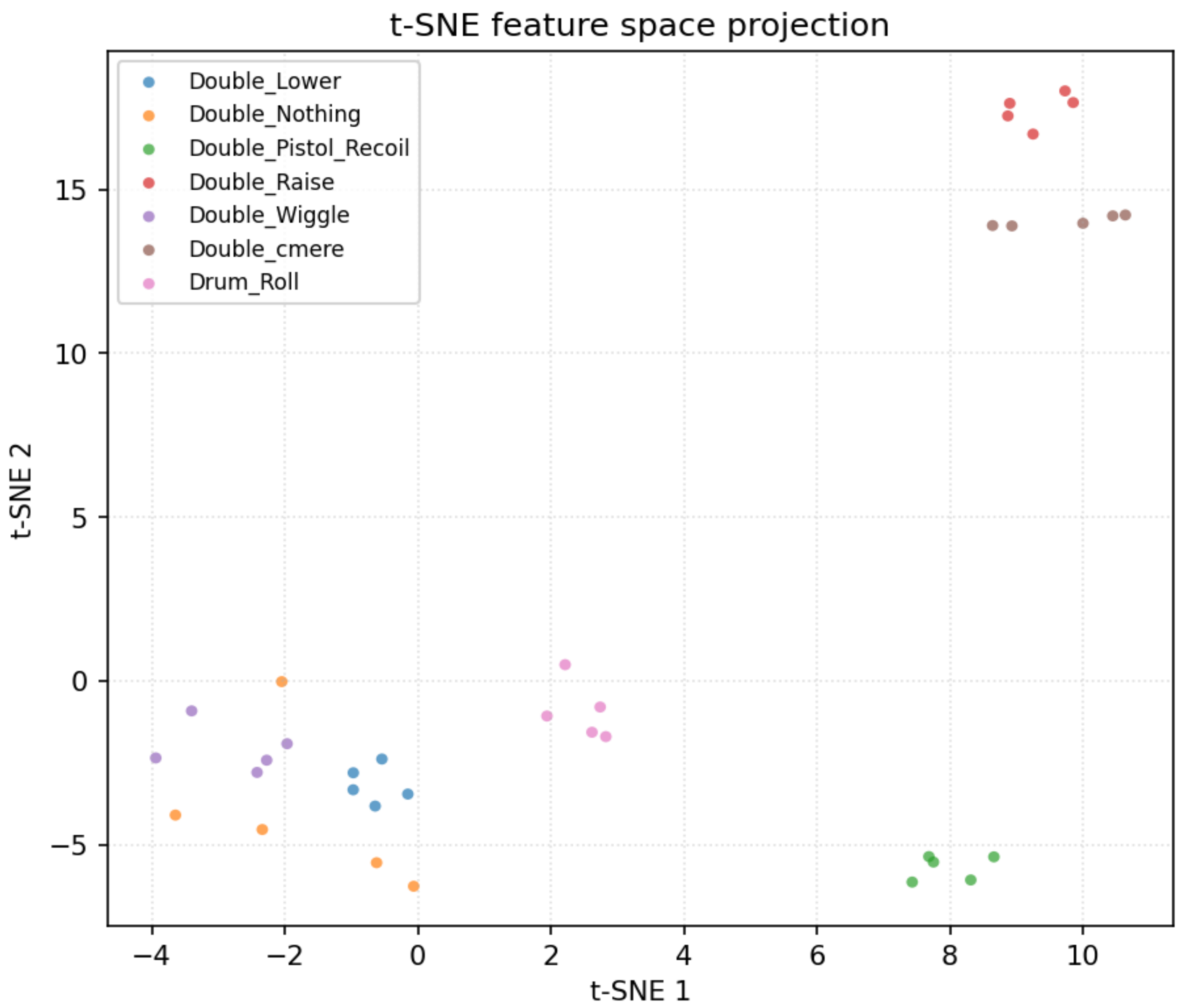
t-SNE feature space projection
150
100
50
0
−50
−100
t-SNE 2
−100
−50
0
50
100
150
t-SNE 1
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

Data diagnostics — user 18_Bridgette

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

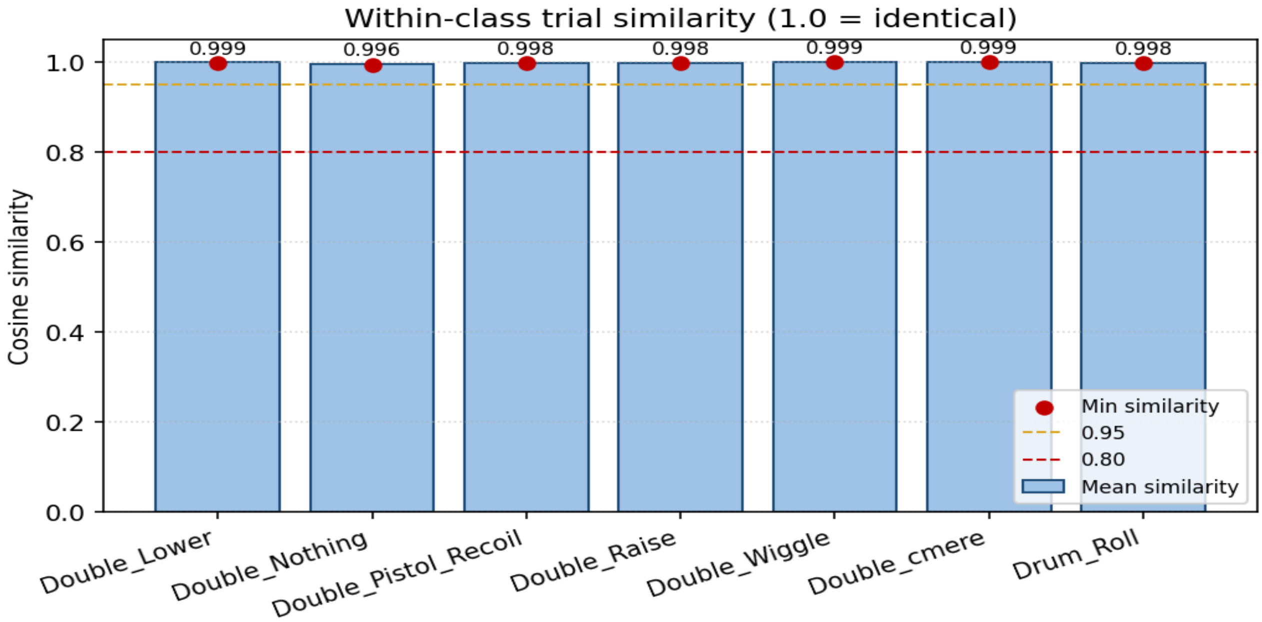


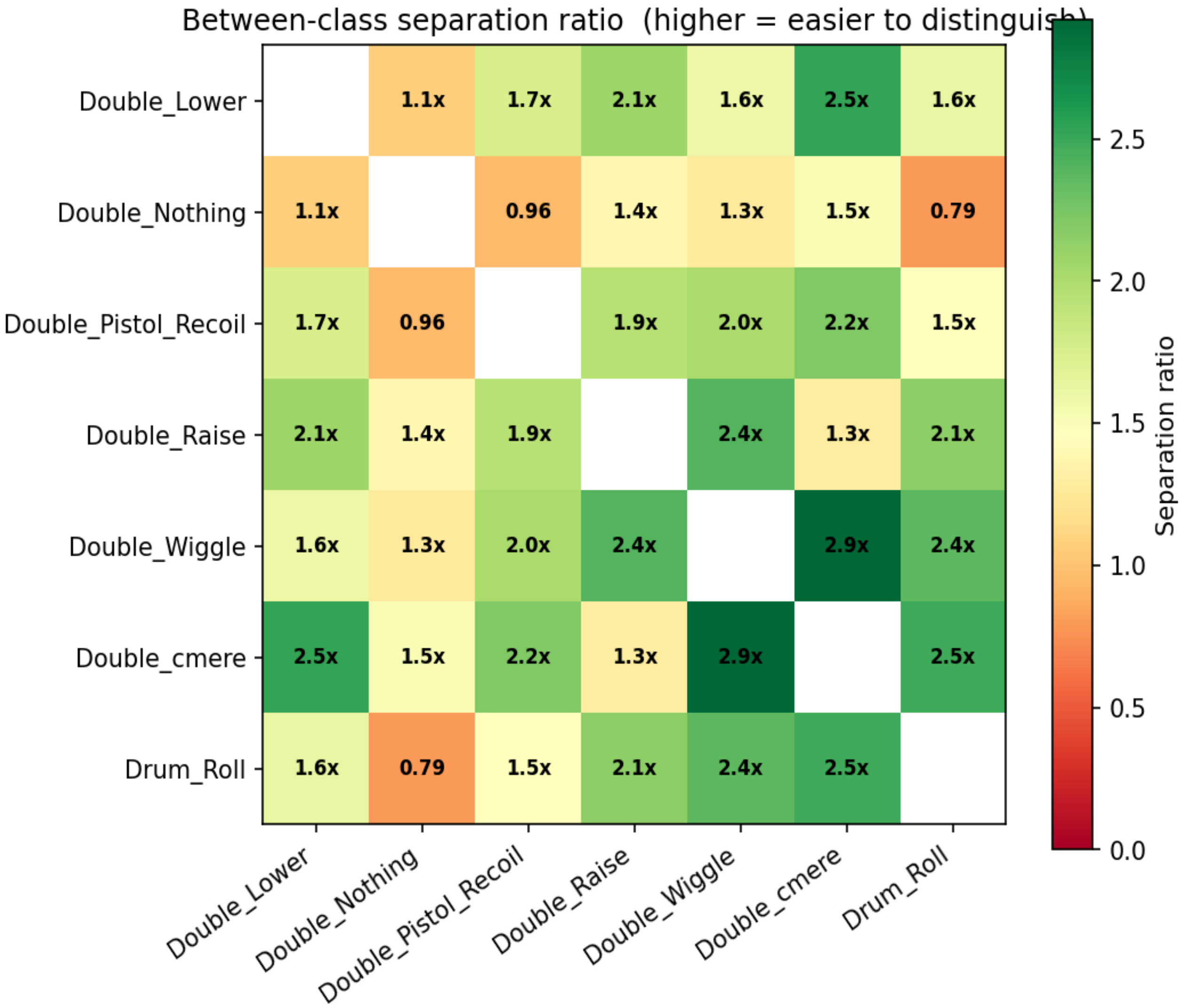


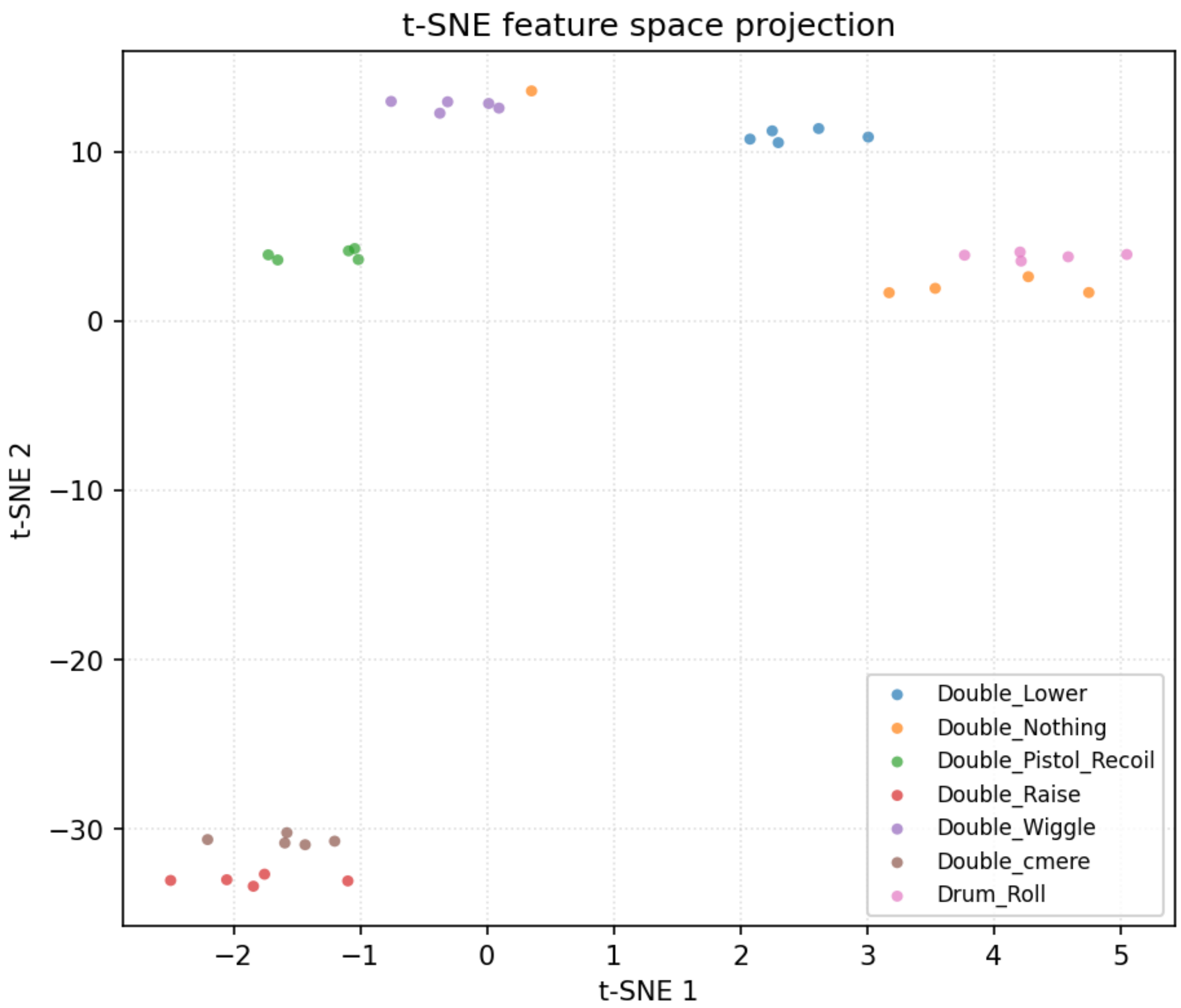

t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
15
10
5
0
−5
−4
−2
0
2
4
6
8
10

Data diagnostics — user 1_Alan

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

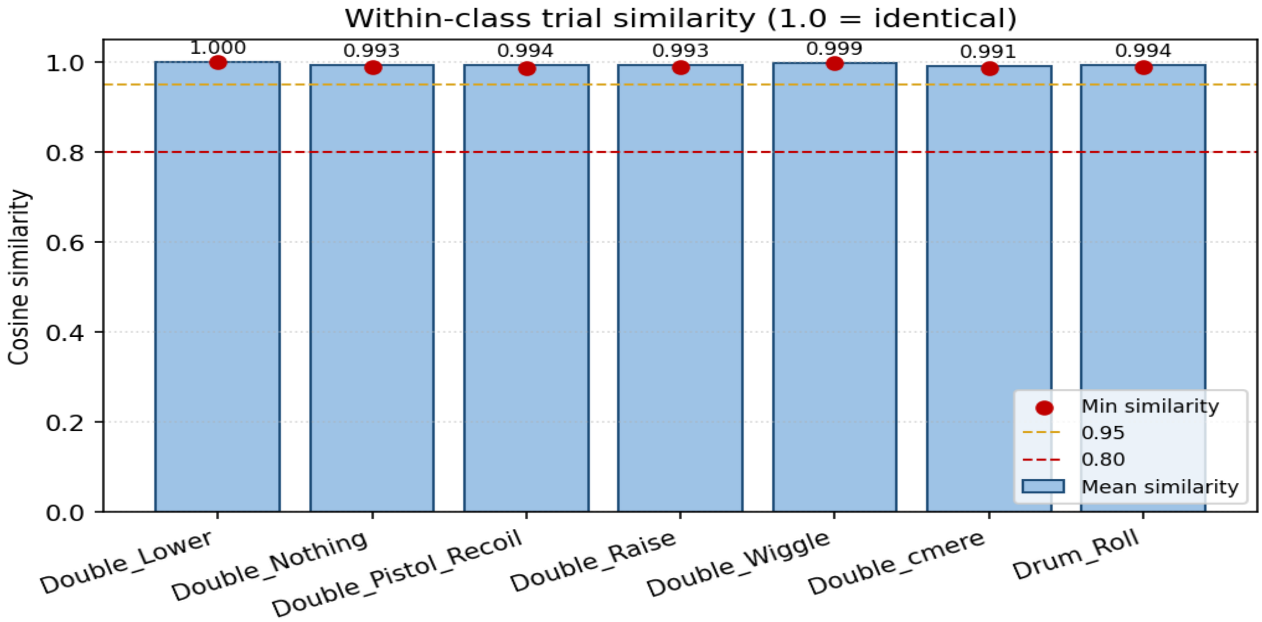


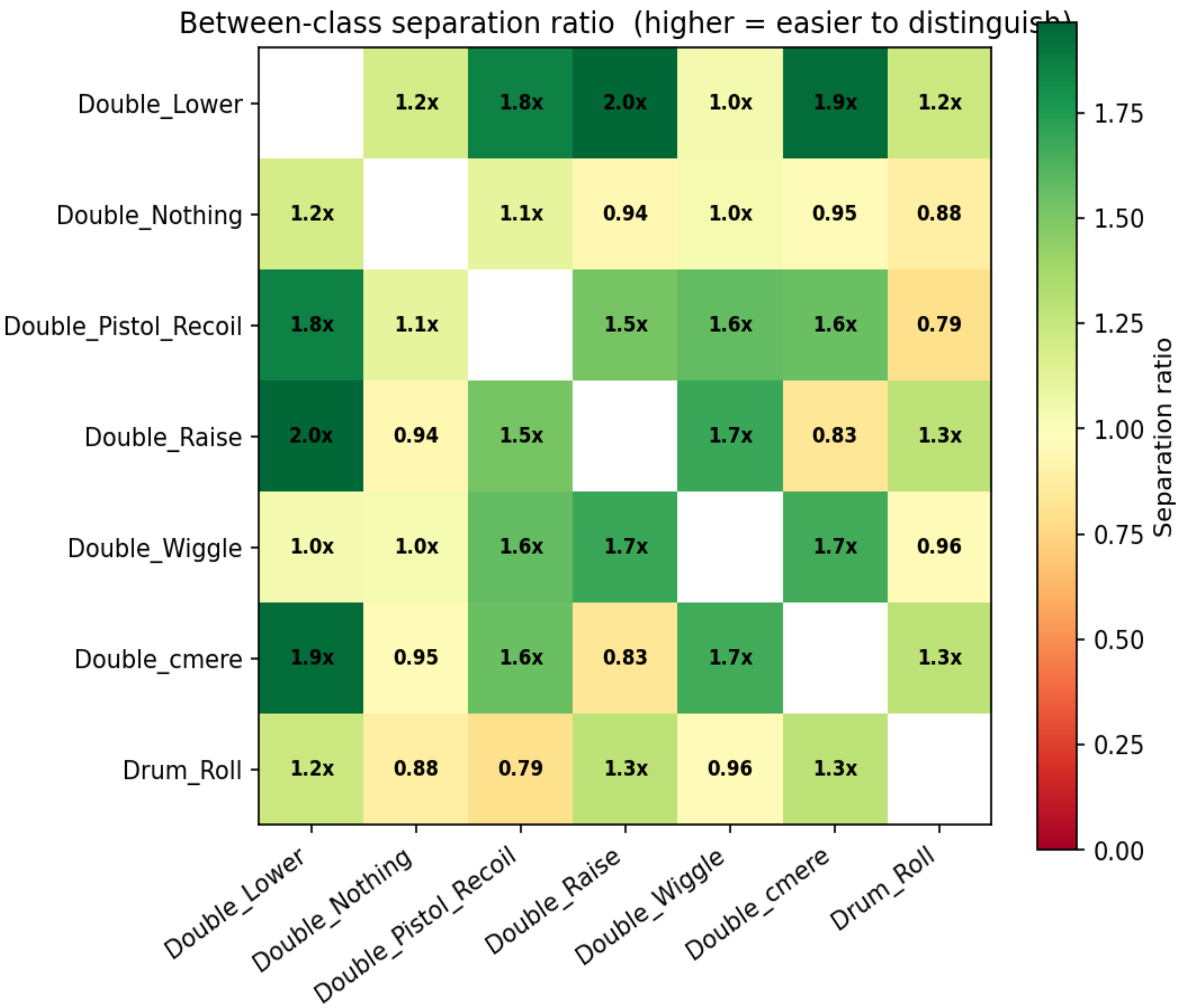


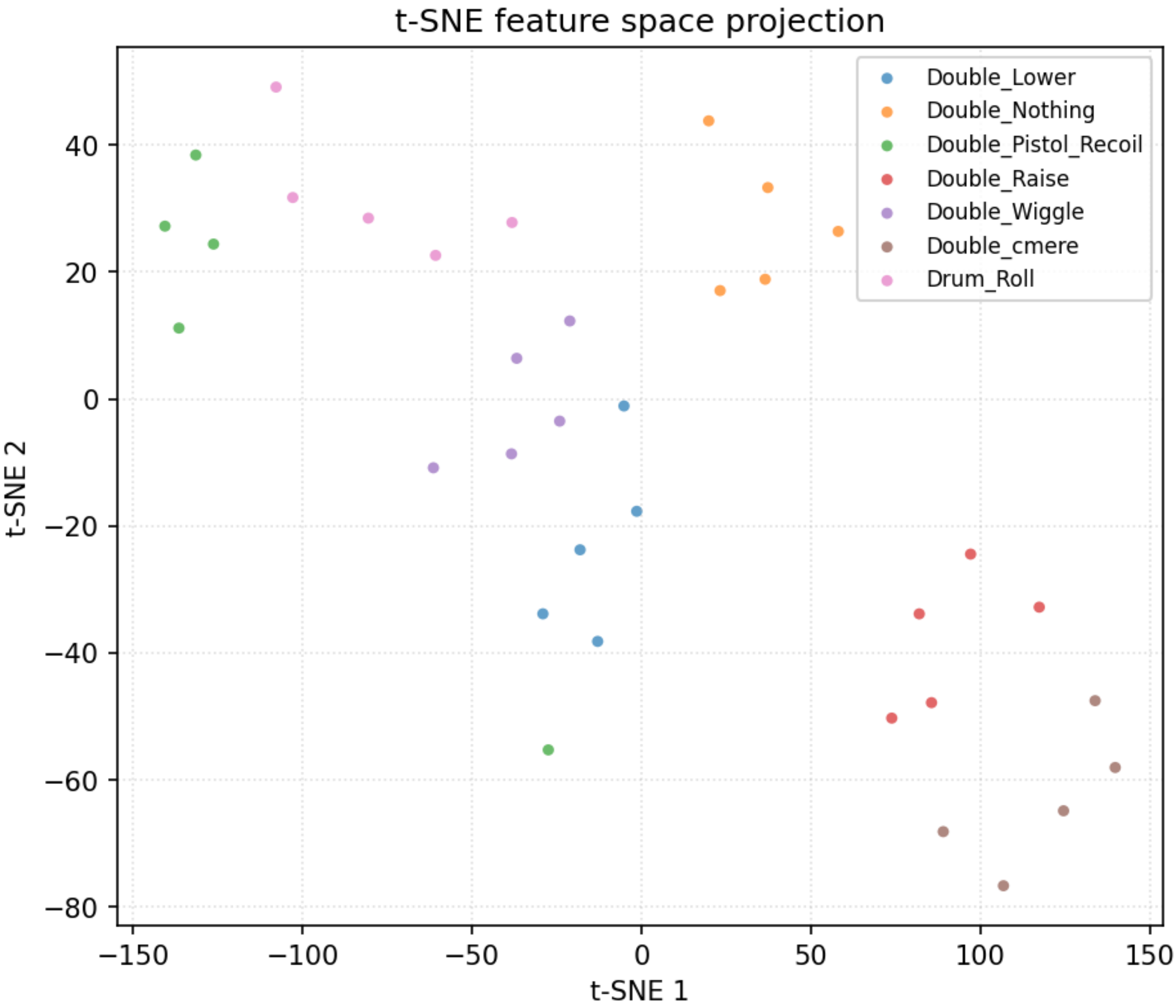

t-SNE feature space projection
t-SNE 2
t-SNE 1
10
0
−10
−20
−30
−2
−1
0
1
2
3
4
5
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

Data diagnostics — user 2_Alex

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

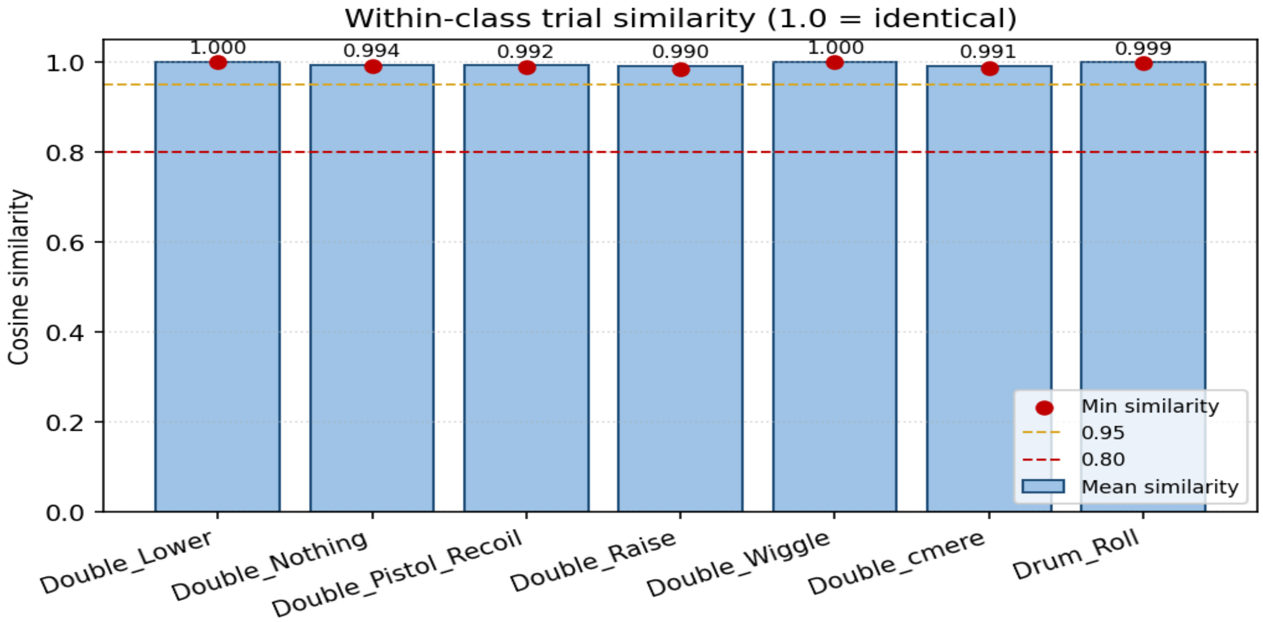


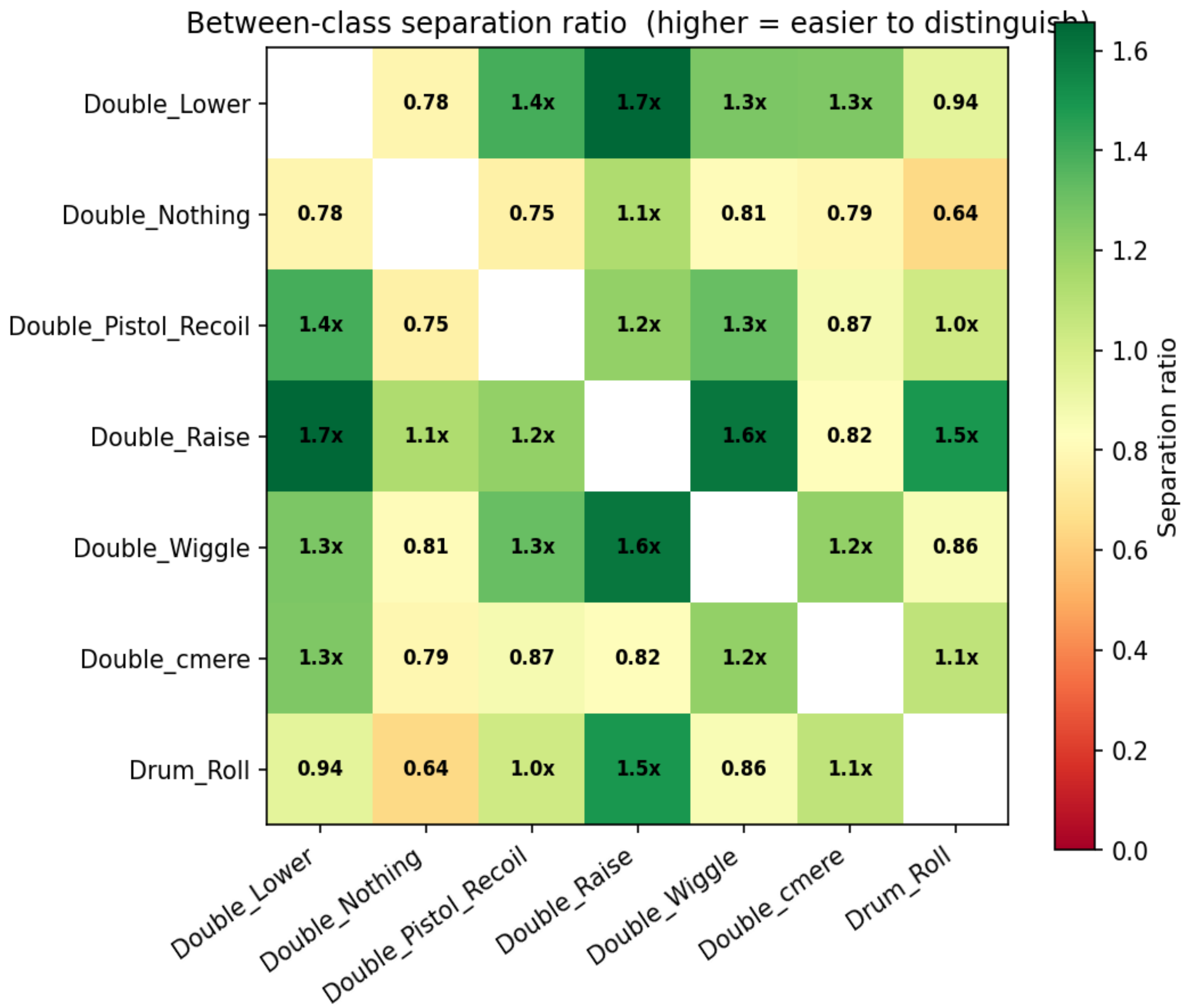


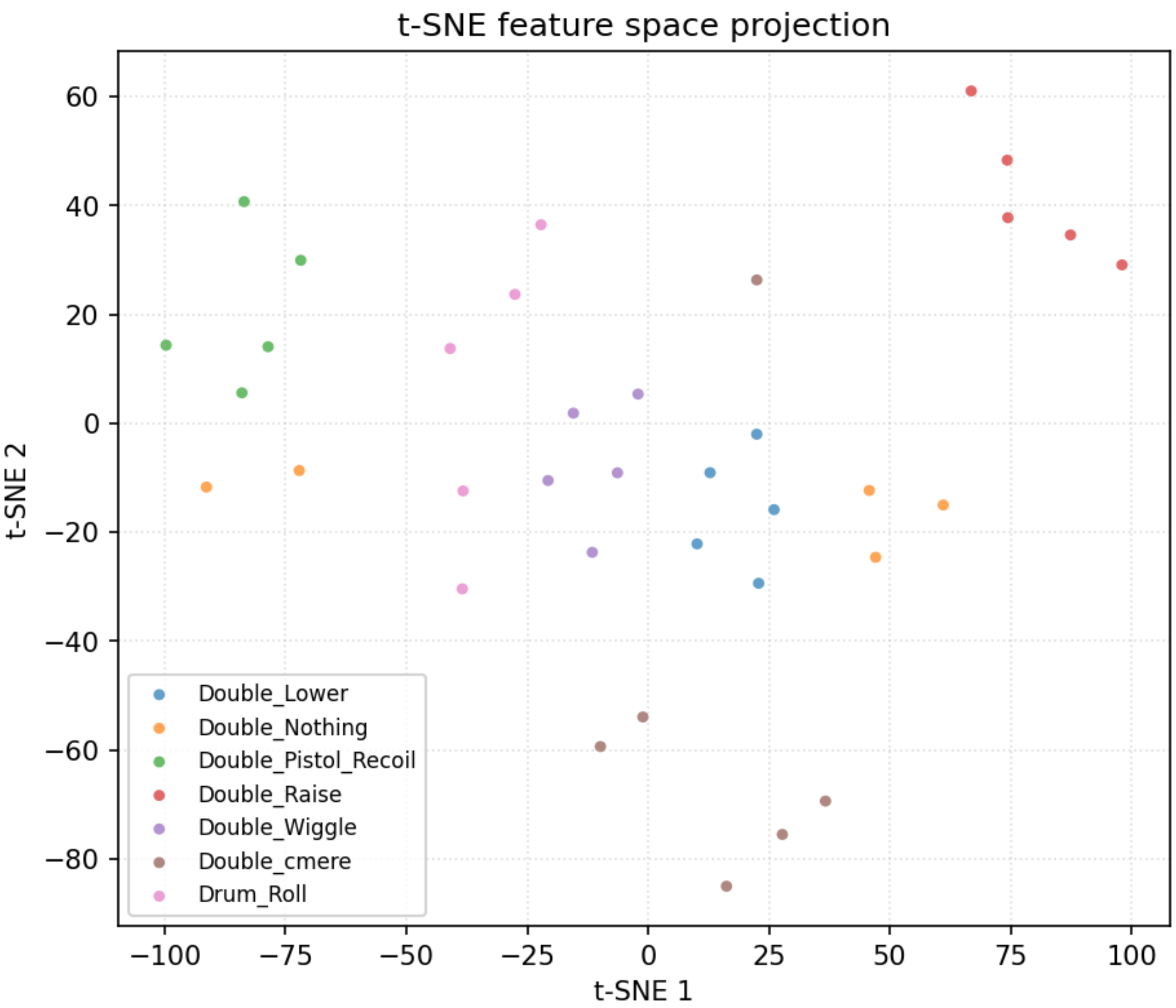

t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150

Data diagnostics — user 3_Anghad

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

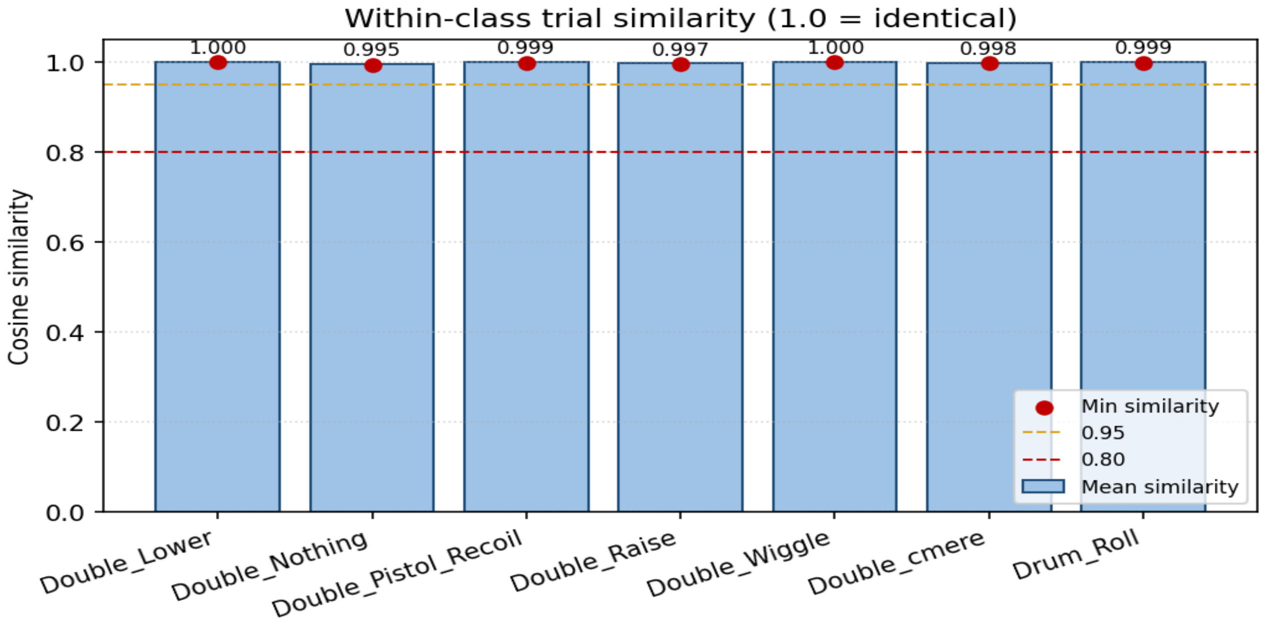


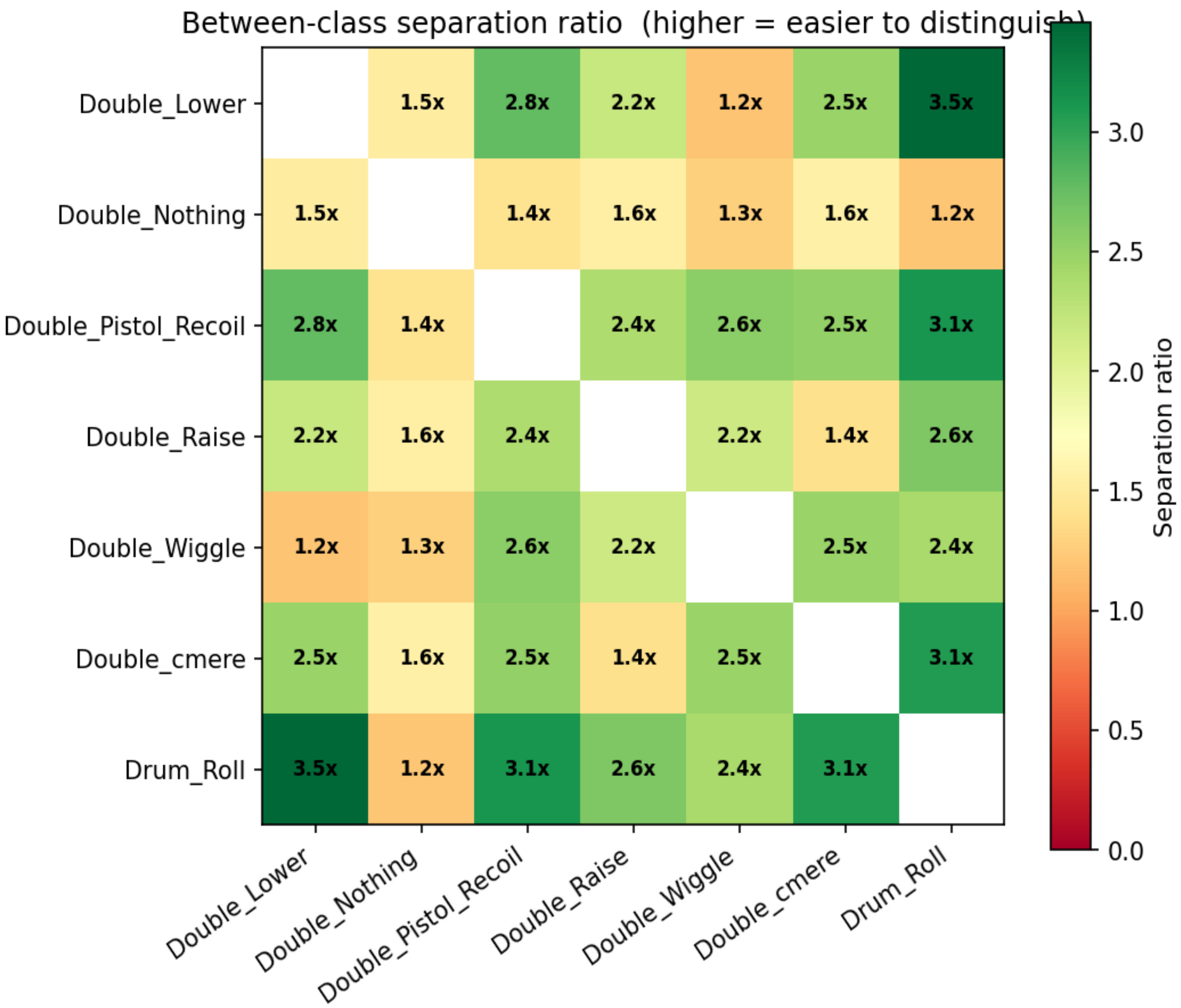



t-SNE feature space projection
t-SNE 2
t-SNE 1
60
40
20
0
−20
−40
−60
−80
−100
−75
−50
−25
0
25
50
75
100
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

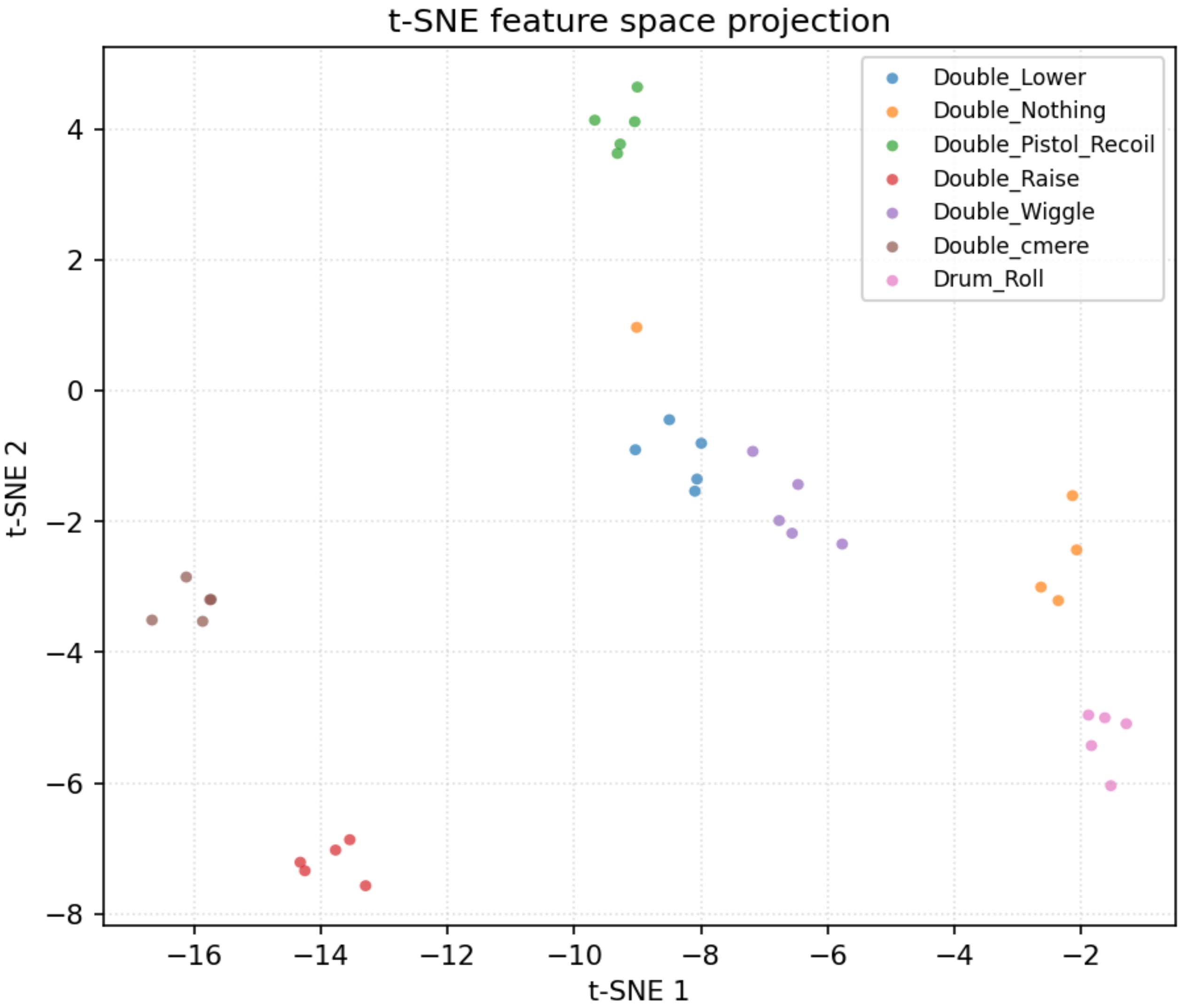
Data diagnostics — user 4_Daniel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).





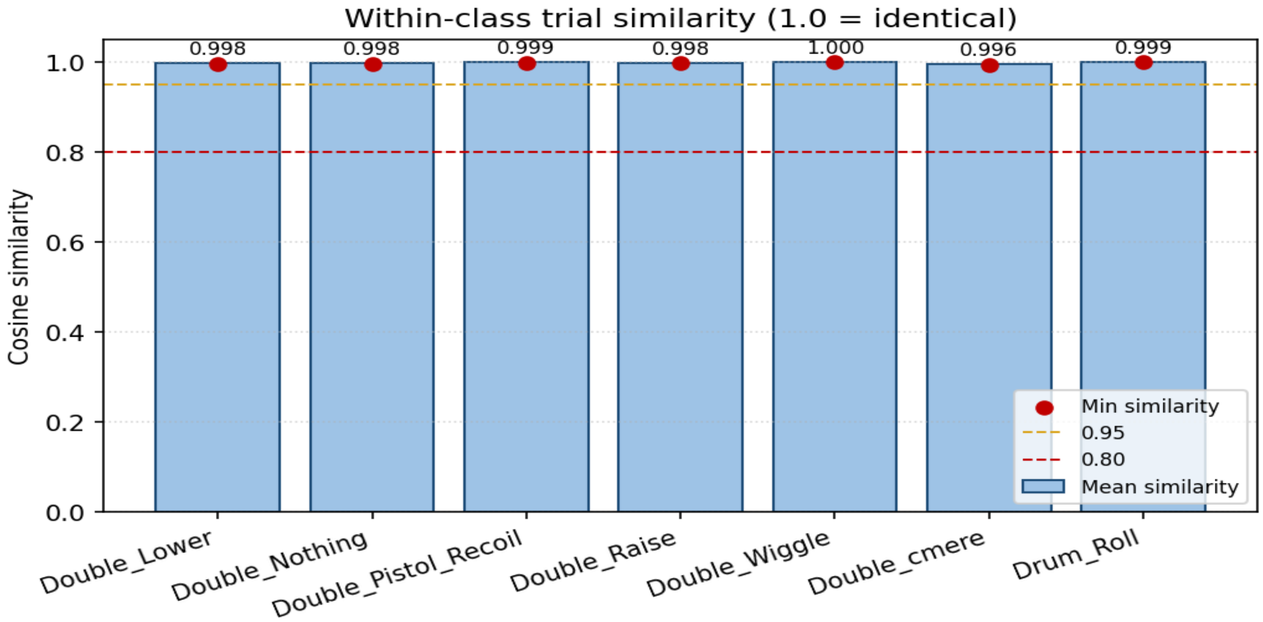
	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		1.5x	2.8x	2.2x	1.2x	2.5x	3.5x
Double_Nothing	1.5x		1.4x	1.6x	1.3x	1.6x	1.2x
Double_Pistol_Recoil	2.8x	1.4x		2.4x	2.6x	2.5x	3.1x
Double_Raise	2.2x	1.6x	2.4x		2.2x	1.4x	2.6x
Double_Wiggle	1.2x	1.3x	2.6x	2.2x		2.5x	2.4x
Double_cmere	2.5x	1.6x	2.5x	1.4x	2.5x		3.1x
Drum_Roll	3.5x	1.2x	3.1x	2.6x	2.4x	3.1x	

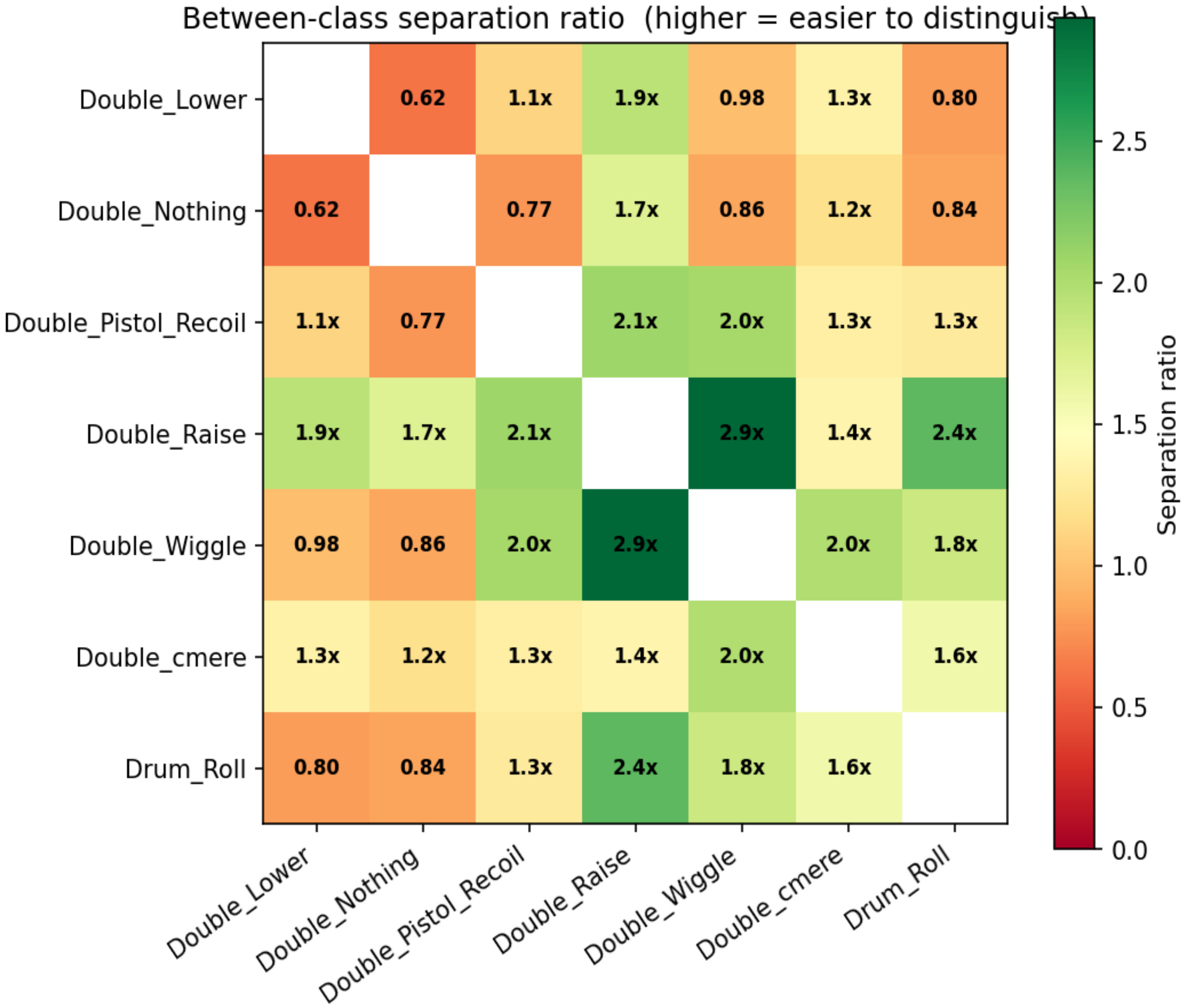


t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 1
t-SNE 2
−16
−14
−12
−10
−8
−6
−4
−2
4
2
0
−2
−4
−6
−8

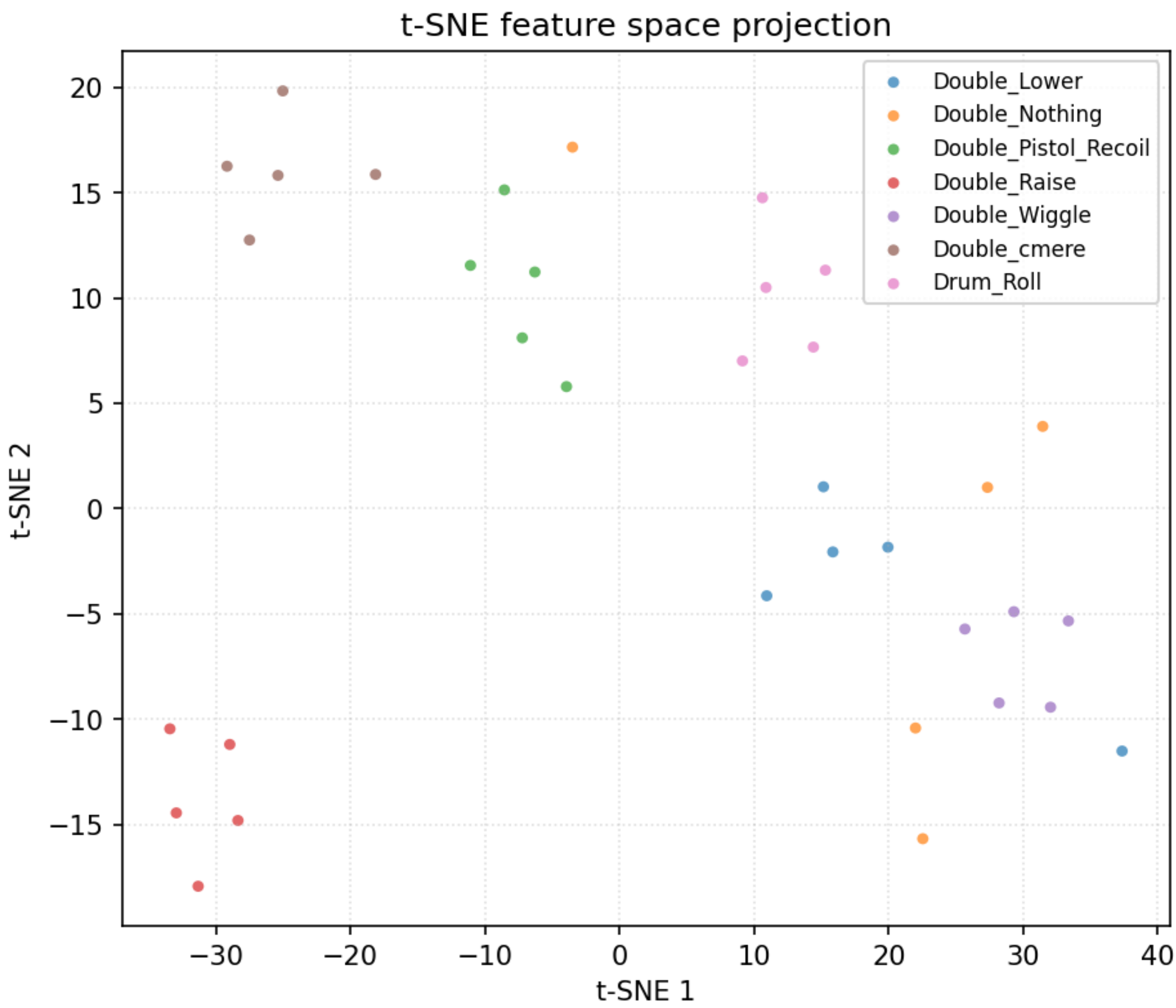
Data diagnostics — user 6_Harry

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).



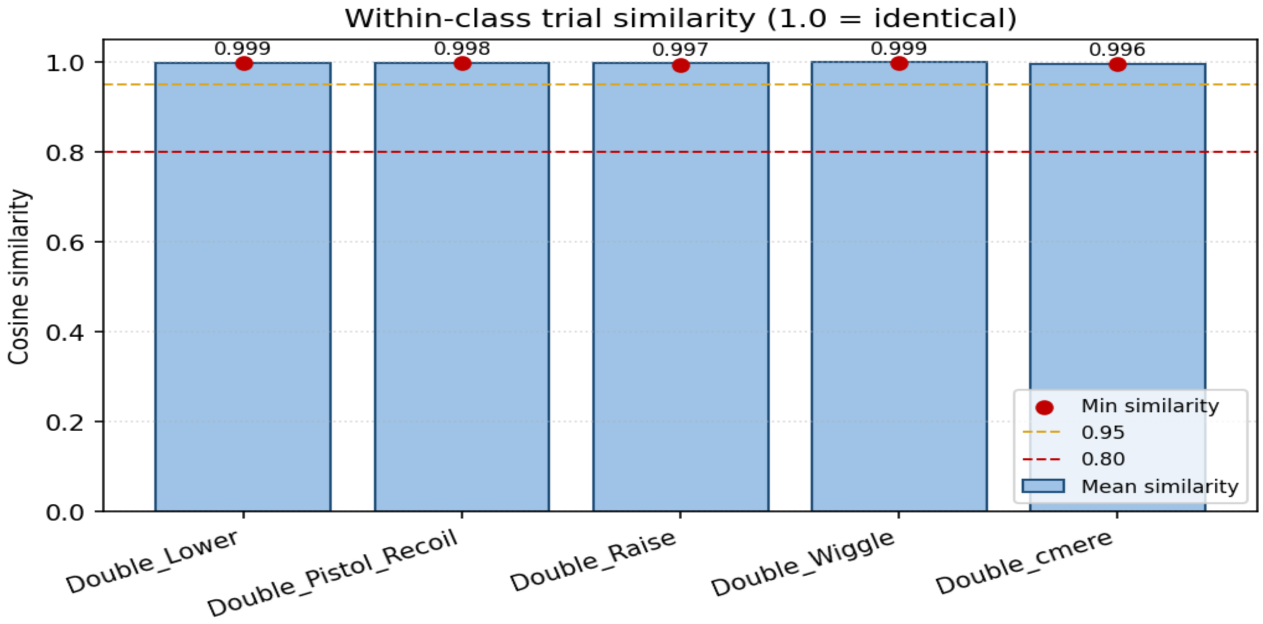


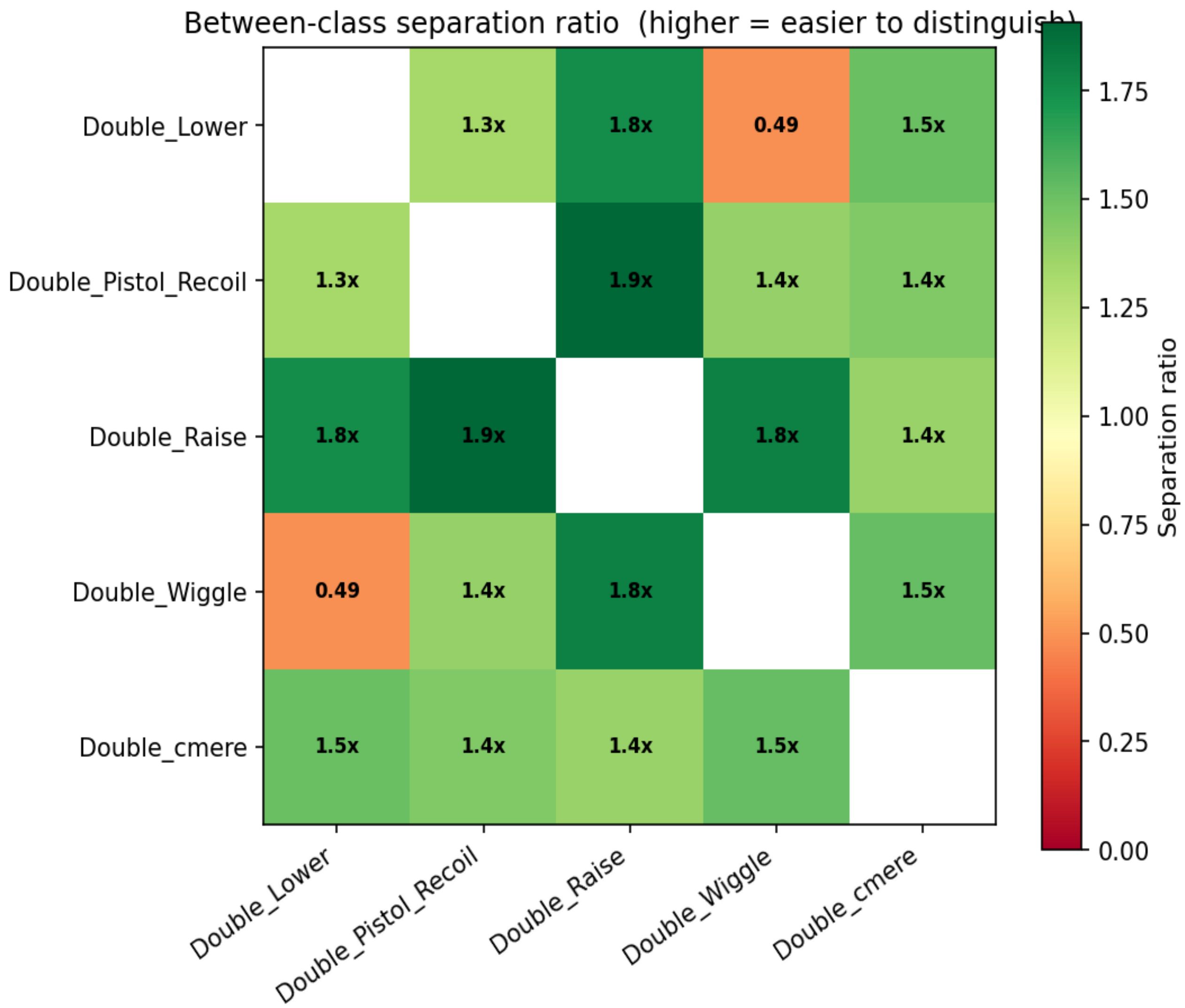
	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		0.62	1.1x	1.9x	0.98	1.3x	0.80
Double_Nothing	0.62		0.77	1.7x	0.86	1.2x	0.84
Double_Pistol_Recoil	1.1x	0.77		2.1x	2.0x	1.3x	1.3x
Double_Raise	1.9x	1.7x	2.1x		2.9x	1.4x	2.4x
Double_Wiggle	0.98	0.86	2.0x	2.9x		2.0x	1.8x
Double_cmere	1.3x	1.2x	1.3x	1.4x	2.0x		1.6x
Drum_Roll	0.80	0.84	1.3x	2.4x	1.8x	1.6x	

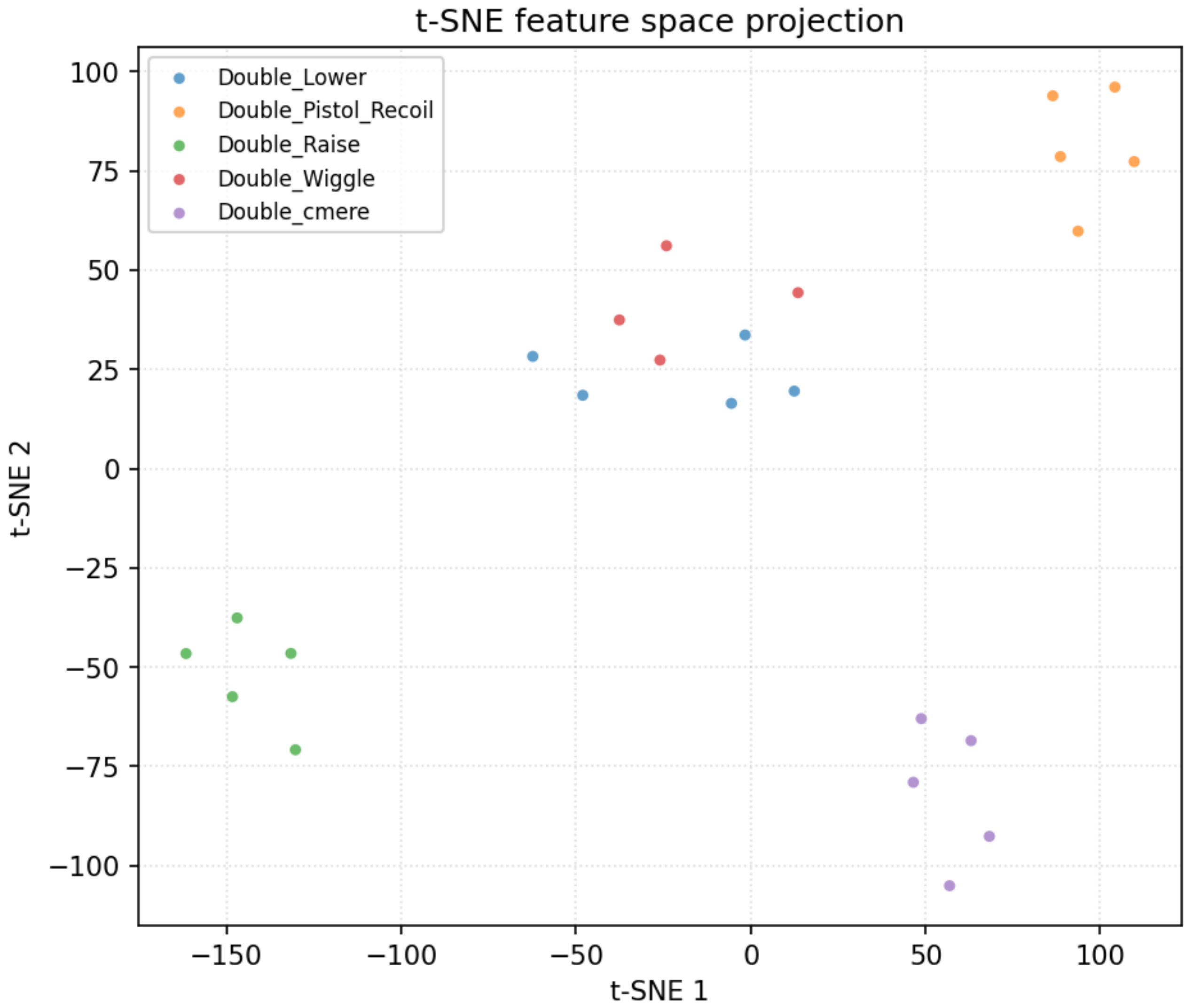

t-SNE feature space projection
Double_Lower
Double_Nothing
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 1
t-SNE 2
20
15
10
5
0
−5
−10
−15
−30
−20
−10
0
10
20
30
40

Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).






t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
100
75
50
25
0
−25
−50
−75
−100
−150
−100
−50
0
50
100
t-SNE 1
t-SNE 2

Variant: BN_GAP_Wide

Conv1D(32,k=8)+BN→Conv1D(64,k=8)+BN→Conv1D(128,k=5)+BN→GAP→Dropout(0.3)→Softmax [LOUO leader]

Trainable params	104,423
LOUO mean ± std	0.9292 ± 0.0614
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

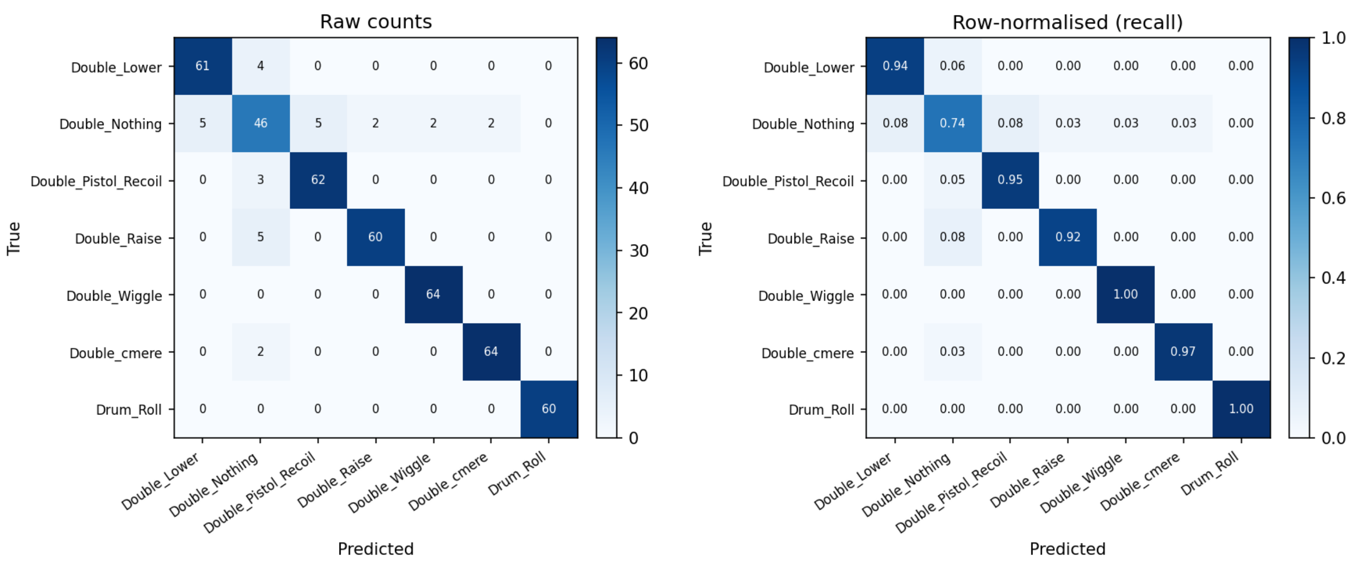
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	377	35	0.8571
2	12_Marcus	14_StephenV2	376	35	0.9714
3	14_StephenV2	15_Tash	376	36	0.9722
4	15_Tash	16_Bella	376	35	1.0000
5	16_Bella	17_Raquel	376	36	0.9444
6	17_Raquel	18_Bridgette	377	35	1.0000
7	18_Bridgette	1_Alan	377	35	0.9143
8	1_Alan	2_Alex	377	35	0.8571
9	2_Alex	3_Anghad	376	35	0.9143
10	3_Anghad	4_Daniel	376	36	1.0000
11	4_Daniel	6_Harry	377	35	0.9143
12	6_Harry	8_Jestin	388	35	0.9429
13	8_Jestin	11_Mansh	388	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.924	0.938	0.931	65
Double_Nothing	0.767	0.742	0.754	62
Double_Pistol_Recoil	0.925	0.954	0.939	65
Double_Raise	0.968	0.923	0.945	65
Double_Wiggle	0.970	1.000	0.985	64
Double_cmere	0.970	0.970	0.970	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.932	0.932	0.932	447
weighted avg	0.932	0.933	0.932	447

Confusion matrix (aggregated across folds)



Final retrain (deployment model)

Variant	BN_GAP_Wide
Held-out user	— (trained on all users)
Held-out accuracy	—
Training-set size	447 samples (447 before augmentation)
Trainable params	104,423
Final train loss	0.0009
Final train accuracy	1.0000
Model path	— (not saved)
Labels path	— (not saved)
Scaler path	— (per-trial norm in use)